Deep Learning 01

MNIST (28×28 크기의 흑백 손글씨 숫자 이미지) 데이터셋

강경림

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Chapter 03

Two_layer_Net 학습 구현

-> MLP(완전 연결층) 기반으로 구현

-> 훈련 데이터, 테스트 데이터: 60000개, 10000개

-> 반복 훈련횟수 : 10000번 >>> 약 167번의 에폭 진행

-> 배치 사이즈: 1000개

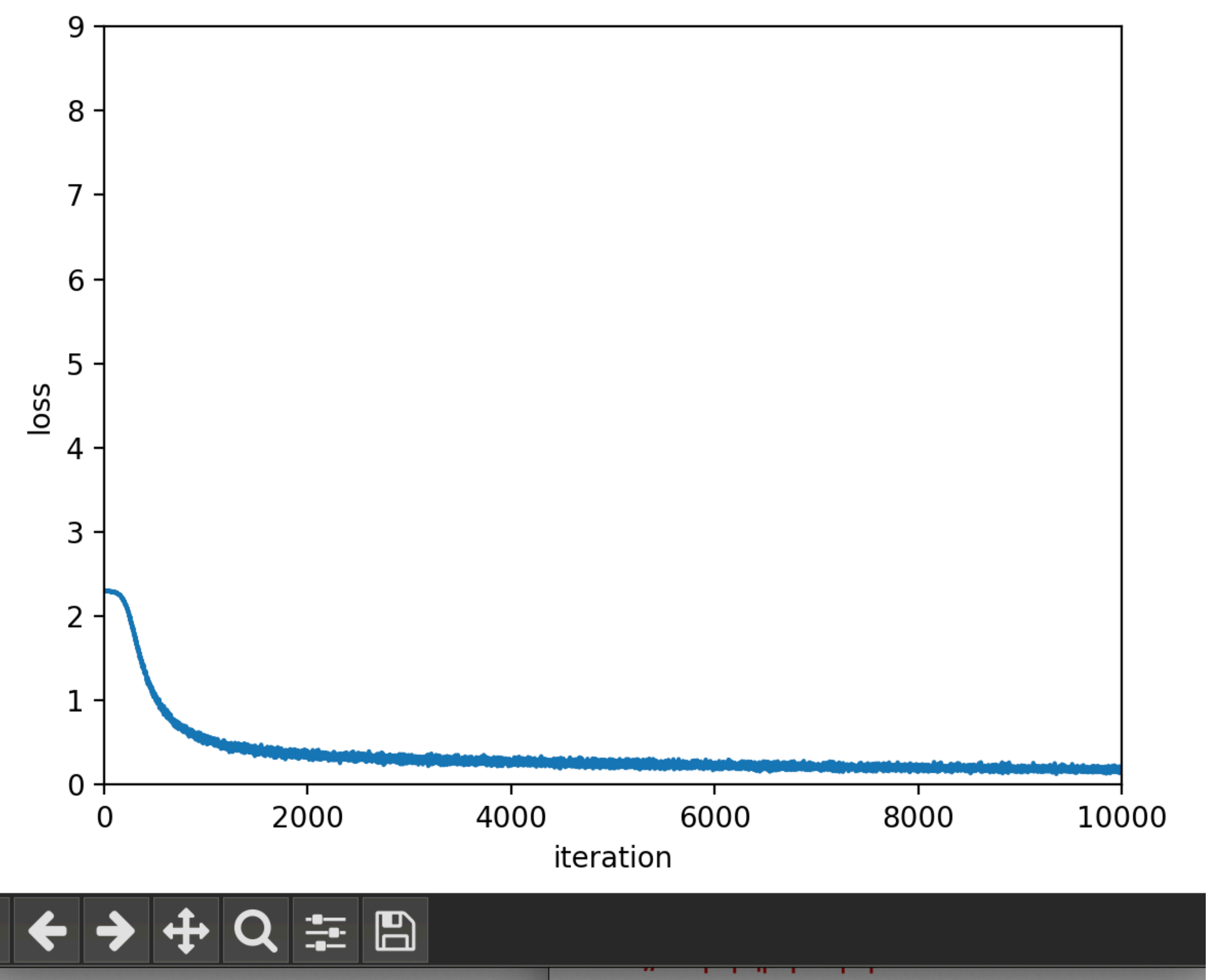
-> 1에폭: 미니 배치학습 60번 진행

-> 입력 데이터 28(28*28=784), 은닉층 50, 출력층 10

Chapter 03

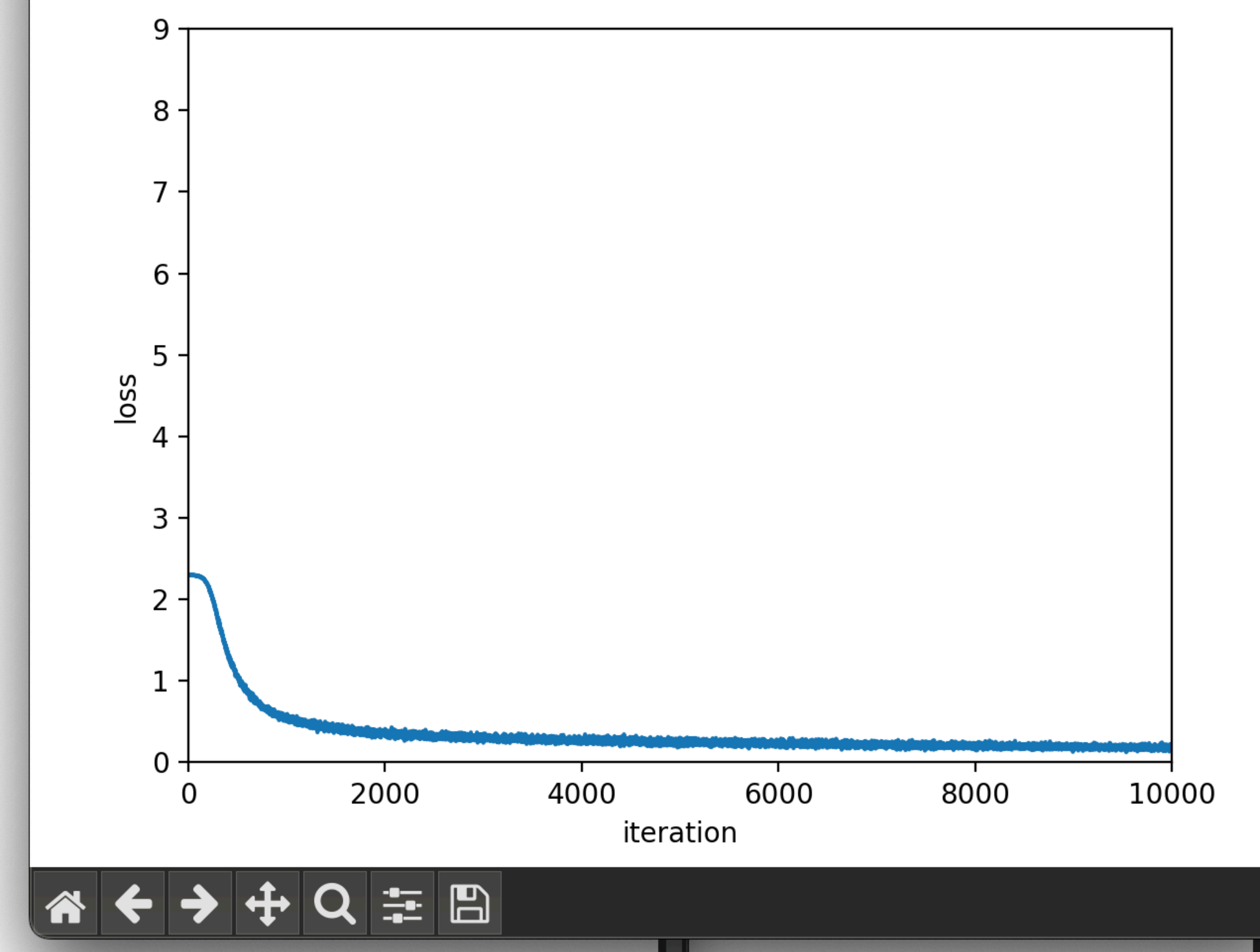
Two_layer_Net 학습 구현

```
train acc, test acc :0.9435166666666667,0.9428
train acc, test acc :0.9437833333333333,0.9426
train acc, test acc :0.9439666666666666,0.943
train acc, test acc :0.9443833333333334,0.9433
train acc, test acc :0.9445333333333333,0.9434
train acc, test acc :0.9447166666666666,0.9439
train acc, test acc :0.9449,0.9438
train acc, test acc :0.9449333333333333,0.9442
train acc, test acc :0.945,0.9448
train acc, test acc :0.9450833333333334,0.9443
train acc, test acc :0.9456,0.9442
train acc, test acc :0.9455666666666667,0.9448
train acc, test acc :0.9457666666666666,0.9443
train acc, test acc :0.94585,0.9444
train acc, test acc :0.9462833333333334,0.9451
train acc, test acc :0.9468666666666666,0.9452
train acc, test acc :0.9471,0.9457
train acc, test acc :0.9466666666666667,0.9449
train acc, test acc :0.9468833333333333,0.9455
train acc, test acc :0.9470833333333334,0.9456
train acc, test acc :0.9473,0.9456
train acc, test acc :0.9475833333333333,0.9461
train acc, test acc :0.94775,0.9458
train acc, test acc :0.9476166666666667,0.9459
train acc, test acc :0.9479166666666666,0.9458
train acc, test acc :0.9482,0.9466
train acc, test acc :0.9482166666666667,0.9458
Training Time: 1m 44s
```




```
batch_mask = np.random.choic
x_batch = x_train[batch_mask
```

```
train acc, test acc :0.9437166666666666,0.9441
train acc, test acc :0.9436,0.9438
train acc, test acc :0.9438666666666666,0.9448
train acc, test acc :0.9437833333333333,0.9443
train acc, test acc :0.9444166666666667,0.9452
train acc, test acc :0.9444,0.9446
train acc, test acc :0.9445333333333333,0.9443
train acc, test acc :0.9446333333333333,0.9454
train acc, test acc :0.9450333333333333,0.9449
train acc, test acc :0.9451333333333334,0.9458
train acc, test acc :0.9452,0.9456
train acc, test acc :0.9456,0.9453
train acc, test acc :0.9457666666666666,0.9462
train acc, test acc :0.9459,0.9458
train acc, test acc :0.9461333333333334,0.9459
train acc, test acc :0.94635,0.9464
train acc, test acc :0.9464833333333333,0.9465
train acc, test acc :0.9467,0.9459
train acc, test acc :0.9470833333333334,0.9462
train acc, test acc :0.9473166666666667,0.9469
train acc, test acc :0.9473166666666667,0.9474
train acc, test acc :0.9476,0.9465
train acc, test acc :0.9477166666666667,0.9478
train acc, test acc :0.9477833333333333,0.9477
train acc, test acc :0.94785,0.9479
train acc, test acc :0.9483166666666667,0.9479
Training Time: 1m 45s
```



-> 교재 결과 소요시간: 1분44초

-> 정확도: 0.9458

-> 결과 소요시간: 1분 45초

-> 정확도: 0.9479

Chapter 07

CNN 구현

-> 합성곱 기반 딥러닝 모델

-> 훈련 데이터, 테스트 데이터: 60000개, 10000개

-> 반복학습: 20번

-> 배치 사이즈: 100개

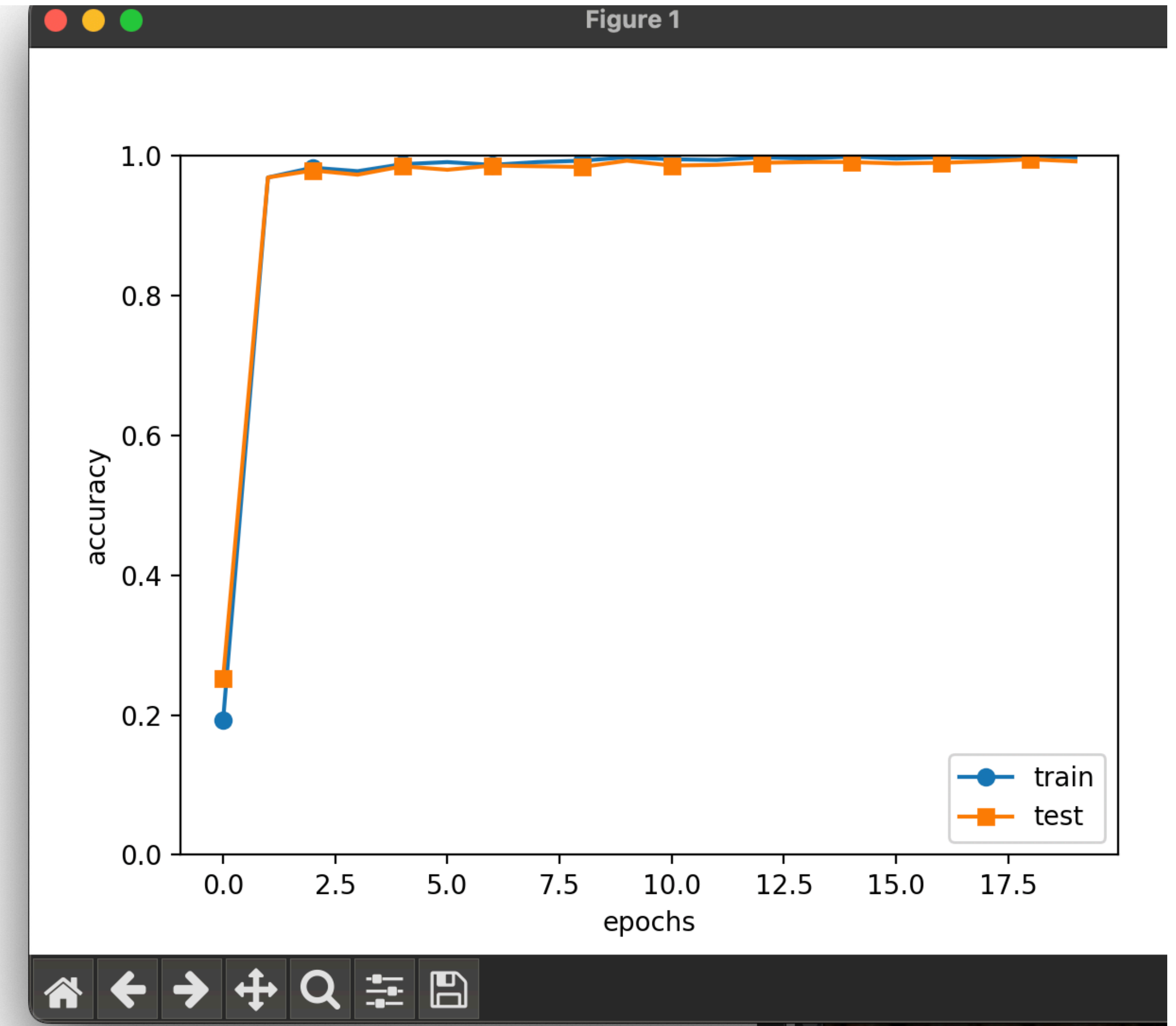
-> 입력 데이터 (1,28,28), 은닉층 100, 출력층 10

-> 필터 개수 30, 필터 사이즈 5, 패딩 0, 스트라이드 1

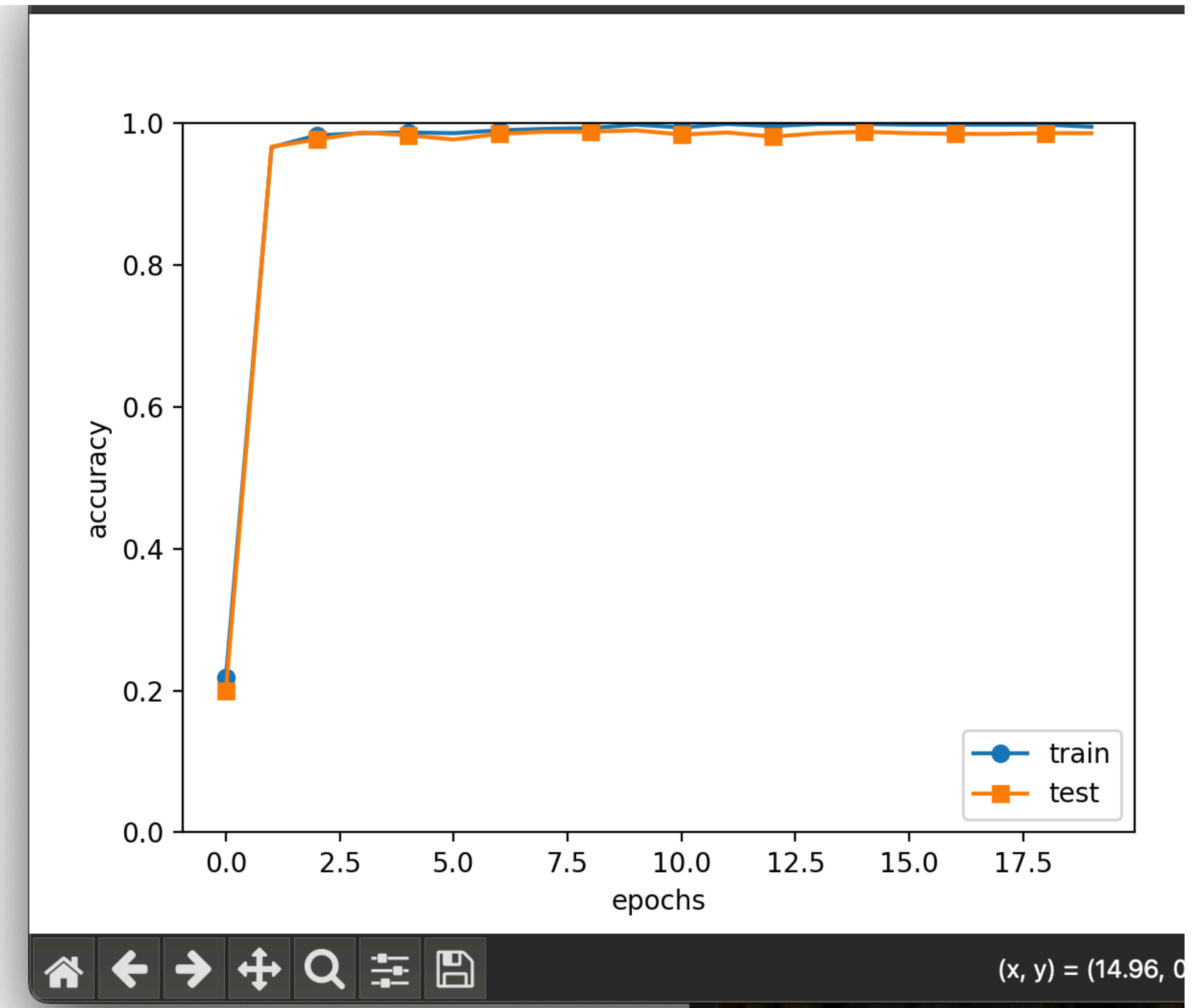
Chapter 07

CNN 구현

```
train loss:0.003212732792827633
train loss:0.0002786411351604109
train loss:0.000619528611962985
train loss:0.001835832403805742
train loss:0.0028917515242366525
train loss:0.0009591041651702717
train loss:0.00019973903530920933
train loss:0.00010985260744904874
train loss:0.0008801047379364749
train loss:0.015829960158102417
train loss:0.0011949703092196378
train loss:0.0009397085804605668
train loss:0.002429111443819706
train loss:0.00016991453637409674
train loss:0.0027147781002138465
train loss:0.00027951537942501754
train loss:8.495783288018703e-05
train loss:0.0027308762616080166
train loss:0.0042646166722302295
train loss:1.2940770932557689e-05
train loss:0.0010162799180679087
train loss:0.0020989488627375257
train loss:0.0024176039032071054
train loss:0.0005400606689957232
train loss:0.00012560119422978084
train loss:0.002037760930419288
train loss:0.03044481532340311
train loss:0.0003495459641262788
train loss:0.000544325488194706
train loss:0.008494537393220229
=============== Final Test Accuracy ===============
test acc:0.9876
Training Time: 0h 23m 15s
Saved Network Parameters!
```




```
train loss:2.293152840136794e-05
train loss:0.0014570554857426868
train loss:0.00021185233055239235
train loss:0.00010326559944943503
train loss:0.0004161701675067185
train loss:0.00036014133209653345
train loss:3.7802902228296686e-05
train loss:0.0016007626818273146
train loss:0.0017487148201569055
train loss:0.0015863121265120462
train loss:0.0019278415144640265
train loss:0.002412334875216546
train loss:0.0003598838762371995
train loss:8.151836767500963e-05
train loss:0.0002949567079691 56
train loss:0.0010330673964383896
train loss:0.0012155678707883892
train loss:0.001202716412003478
train loss:0.0009237546315528388
train loss:2.440298934220786 3e-05
train loss:0.0001471057783461994
train loss:0.0003314336371140615
train loss:0.0011814240255471062
train loss:0.00023589377413955122
train loss:0.00014467072473241325
train loss:0.0015334890531881976
train loss:0.00036792483650792656
train loss:0.00023323083774648748
train loss:0.001612931544199811
=============== Final Test Accuracy ===============
test acc:0.9894
Training Time: 0h 23m 8s
Saved Network Parameters!
```



-> 교재 결과 정확도: 0.9876

-> 소요시간: 23분 15초

-> 구현 결과 정확도: 0.9894

-> 소요시간: 23분 8초

Chapter 08
심층 신경망 구현

-> 합성곱 계층 + 완전 연결층 딥러닝 모델

-> 네트워크 구조: 6개의 합성곱 계층 + 완전 연결층 (Affine) + 드롭아웃 + 소프트맥스

-> 훈련 데이터, 테스트 데이터: 60000개, 10000개

-> 반복학습: 20번

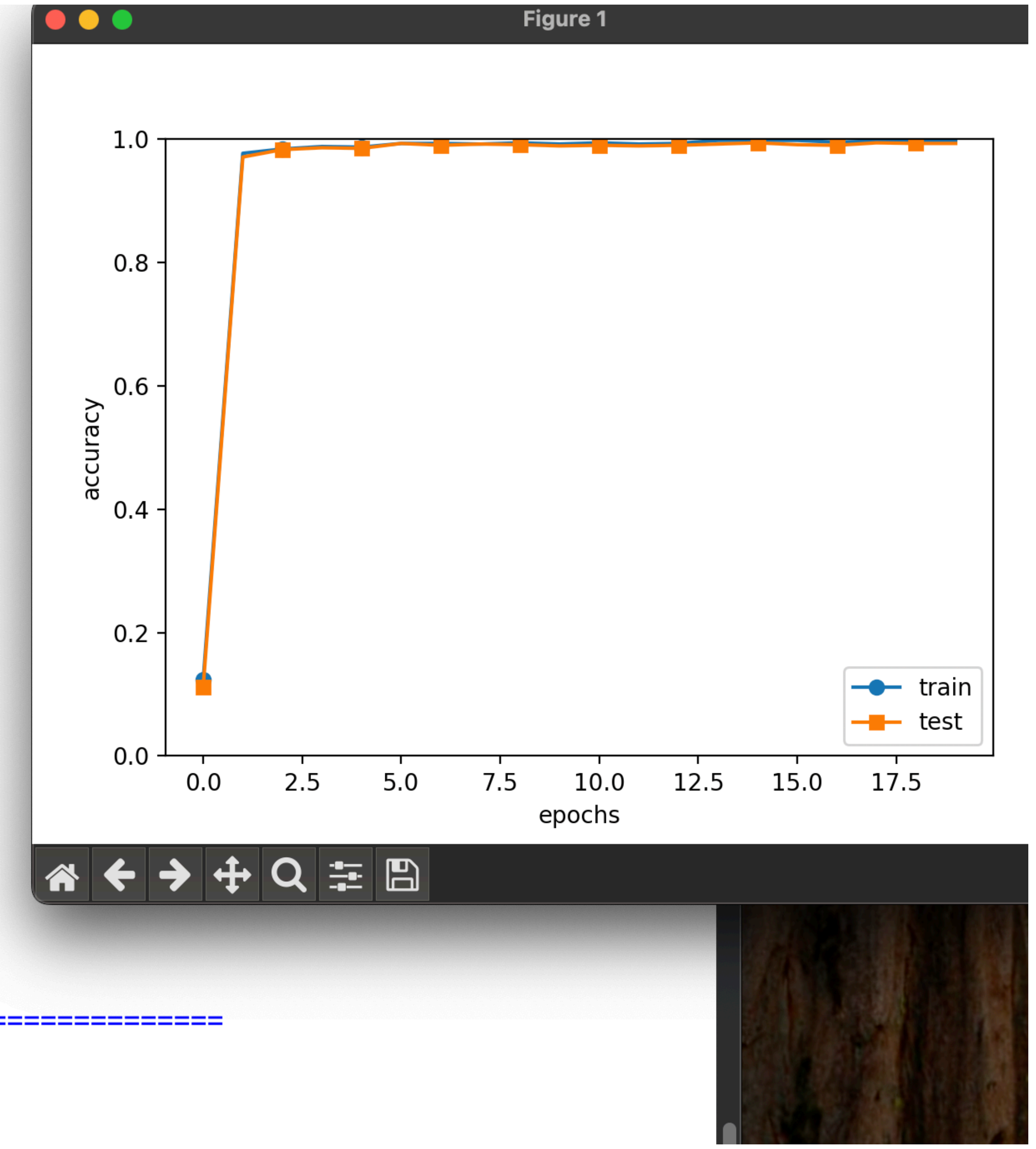
-> 배치 사이즈: 100개

-> 평가 샘플 수(1에포크): 1000개

Chapter 08

심층 신경망 구현

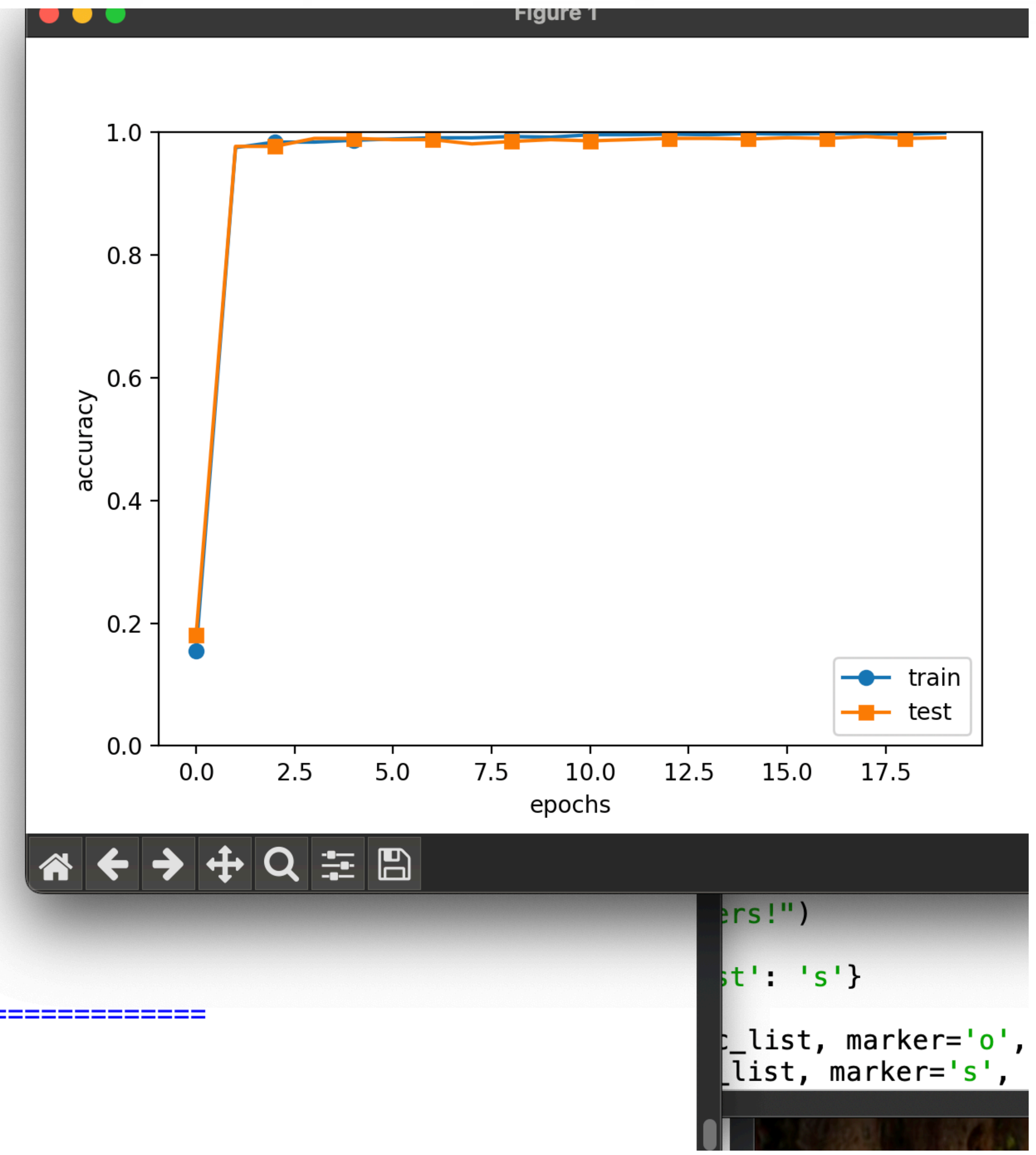

```
train loss:0.9426144757588321
train loss:0.7735305403397597
train loss:0.972521307660231
train loss:1.0131468214074022
train loss:0.872520984460602
train loss:0.7934414344355577
train loss:0.8946524982308055
train loss:0.9427004585488696
train loss:0.7842032972949915
train loss:0.97163411538311
train loss:0.818997875234948
train loss:0.8341111771556322
train loss:0.8938465697577976
train loss:0.8962286231812528
train loss:0.9339796113370668
train loss:0.9843426872514328
train loss:0.783549617984252
train loss:0.9845157135591478
train loss:1.0856430336041643
train loss:0.8760388553581429
train loss:0.8071513881587741
train loss:0.9117088627015322
train loss:0.7187288748219081
train loss:0.8839583647845346
train loss:0.9248462947816287
train loss:0.6745672860834362
train loss:0.9167824144271727
train loss:0.8529746770820346
train loss:1.0334101465447791
train loss:1.027388349950858
train loss:0.9061032612596405
=============== Final Test Accuracy ===============
test acc:0.9947
Training Time: 1h 54m 2s
Saved Network Parameters!
```



-> 교재 결과 소요시간: 1시간 54분 2초

-> 정확도: 0.9947

```
train loss:0.7583293030035674
train loss:0.7323010839619087
train loss:0.7326471704159816
train loss:0.9951956796690362
train loss:0.8965128551796734
train loss:0.8528771668026387
train loss:0.8554518522798732
train loss:0.917597640608992
train loss:0.8203629432006629
train loss:0.8341778631041266
train loss:0.9727210096879391
train loss:0.8575364665737214
train loss:0.7515482965196368
train loss:0.8200133390791424
train loss:0.7495898911414001
train loss:0.897443387037734
train loss:0.8277377707771834
train loss:0.8847736349505338
train loss:0.8152903099444775
train loss:0.8562759039107849
train loss:0.8459132340178673
train loss:0.8545451805551503
train loss:0.9791214839158678
train loss:0.7286957794081426
train loss:0.7963182647903416
train loss:0.8037700199091853
train loss:0.8358795811369285
train loss:0.848963030268565
train loss:0.8278440747903535
train loss:0.8898472299362945
train loss:0.9093309376959702
=============== Final Test Accuracy ===============
test acc:0.9932
Training Time: 1h 53m 34s
Saved Network Parameters!
```



-> 결과 코드 소요시간: 1시간 53분 34초

-> 정확도: 0.9932

CNN(Convolutional Neural Network, 합성곱)

신경망 모델

CNN: 다양한 데이터에서 패턴을 학습하는데 성능이 좋은 신경망 모델

구조: 여러 개의 층을 쌓아서 입력 데이터를 효과적으로 처리할 수 있도록 설계된 구조

특징: 1. 다차원 데이터 처리 가능 -> 1D,2D,3D 등 다양한 형태 다룸

2. 공간 불변성 -> 필터 통해 중요한 특징 학습

3. 계층적 특징 학습: 선 같은 저수준 특징 학습 후 얼굴, 사물 같은 고수준 특징 학습

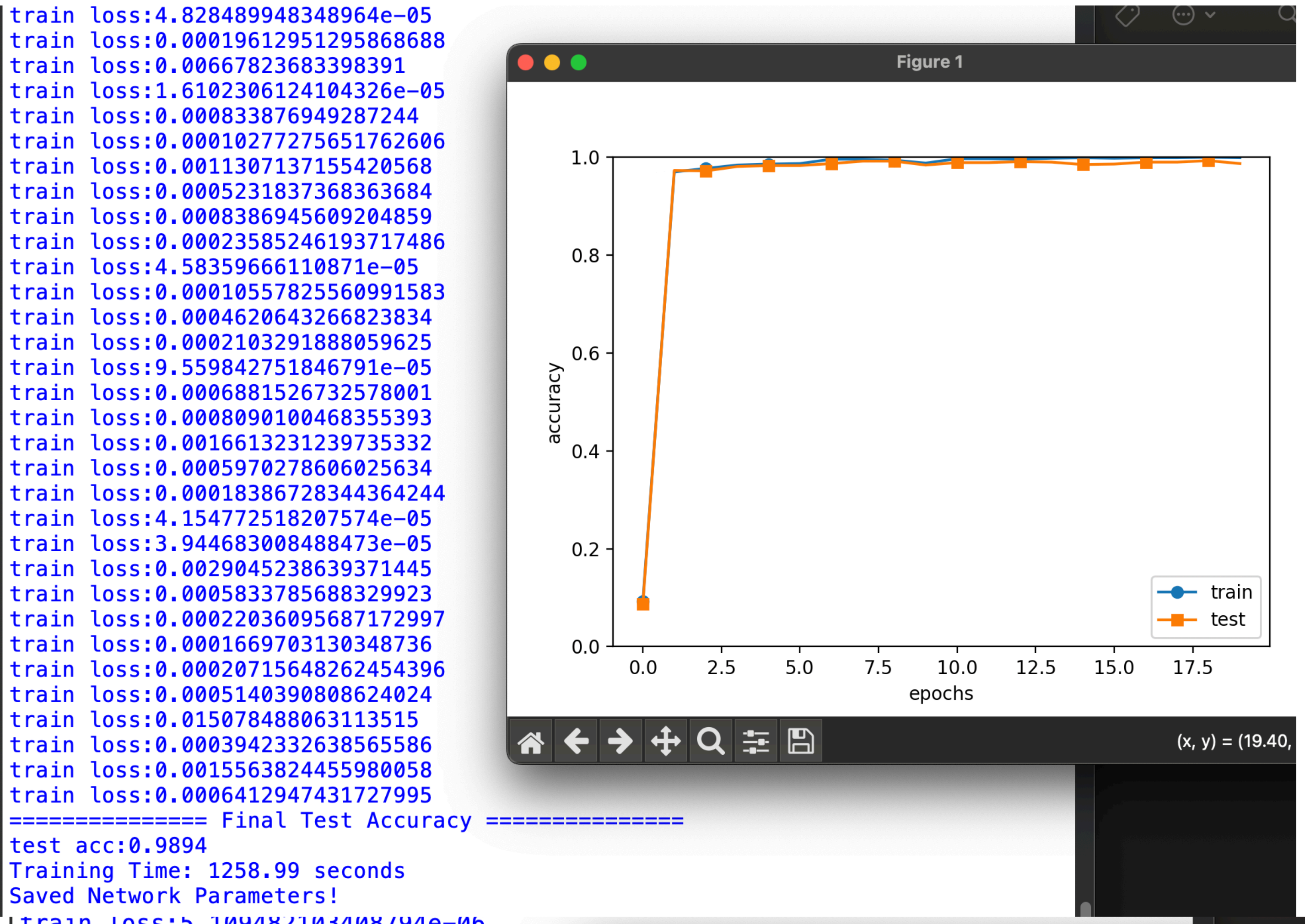
장점: 공간적 특징 보존(이미지 구조 유지), 특징 자동 추출

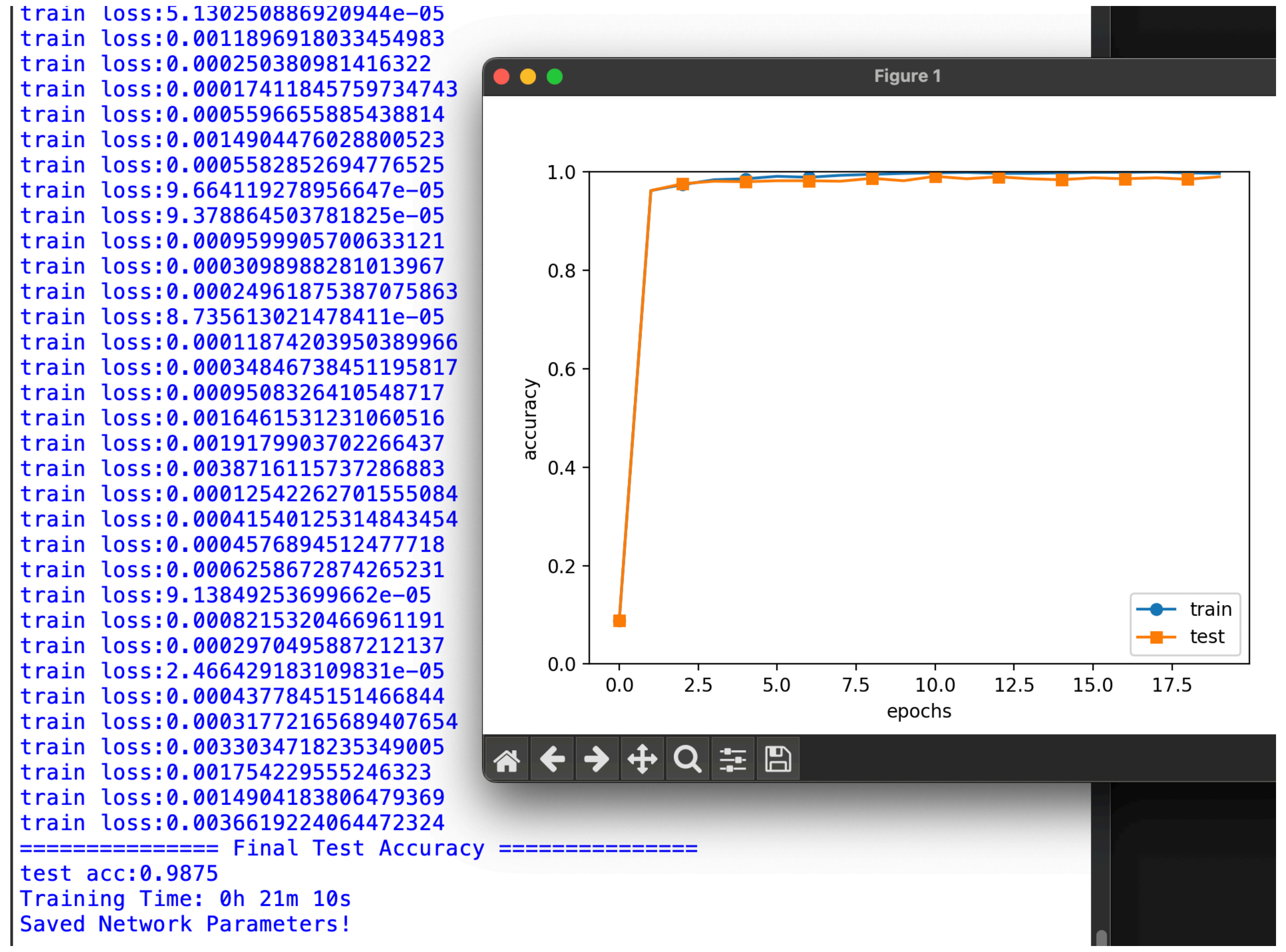
한계: 1. 높은 연산 비용 -> 계산량 많음

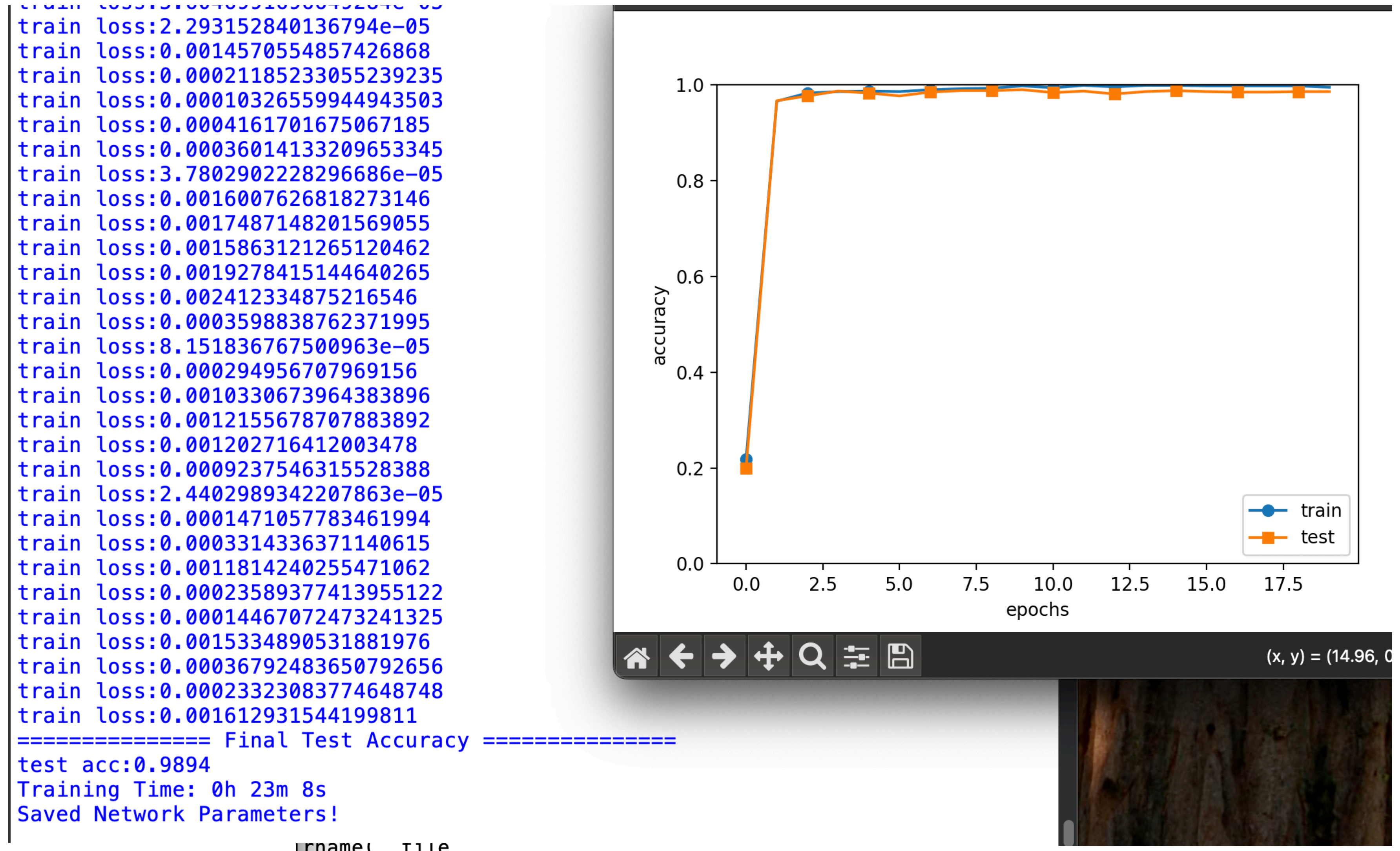
2. 대량의 학습 데이터 필요, 설계 복잡

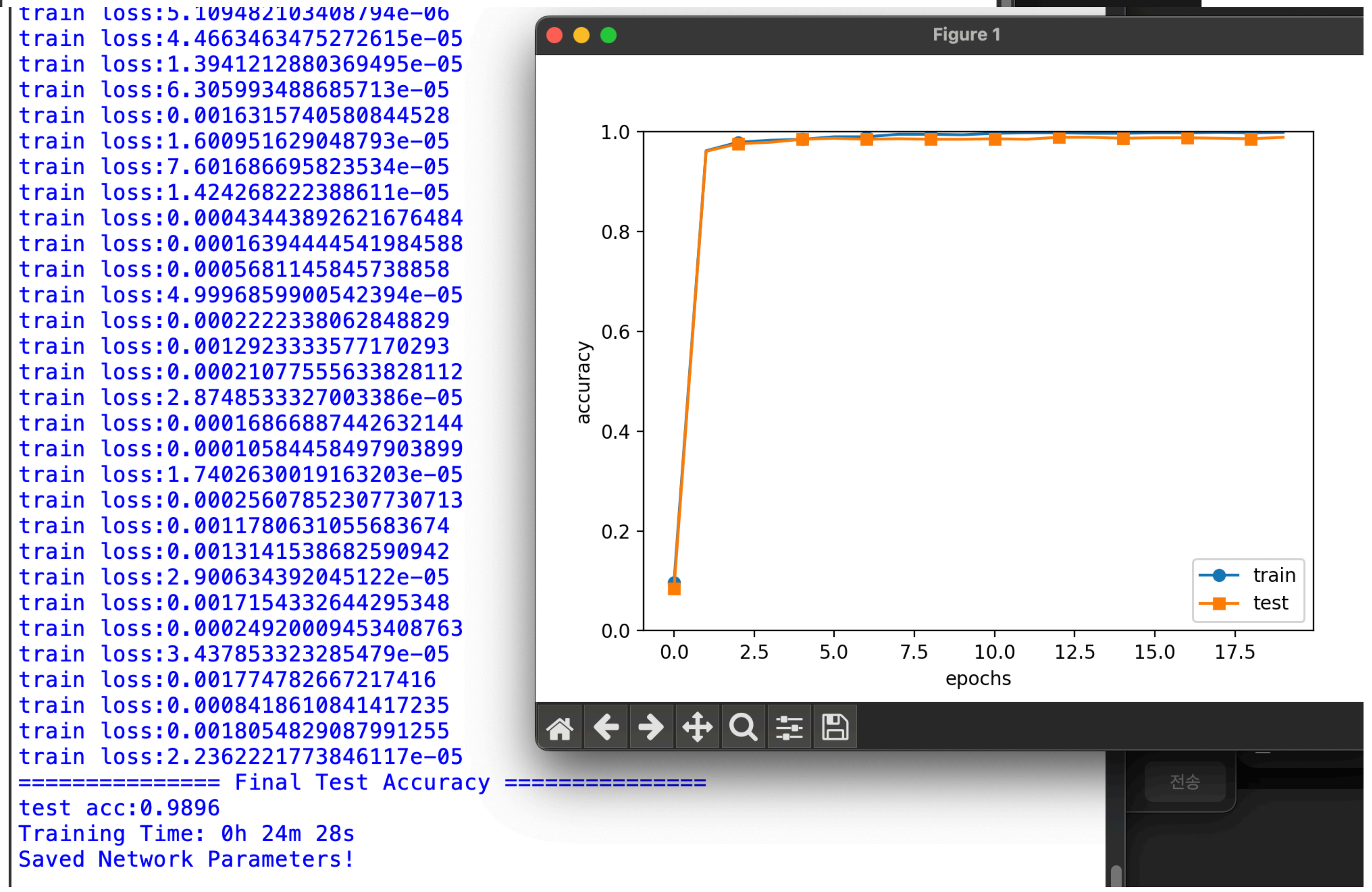
Chapter 07

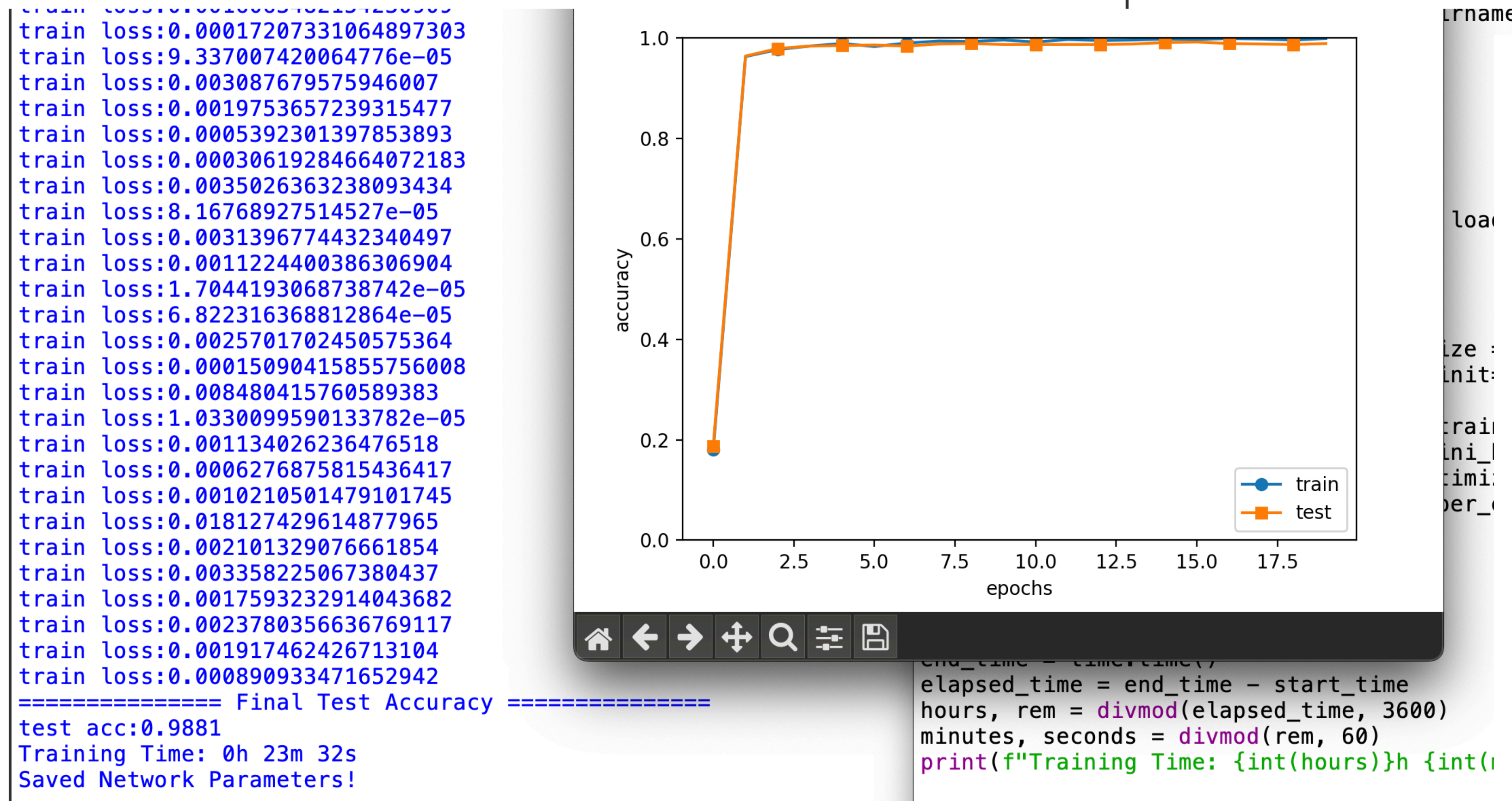
CNN 구현 - 구현 결과

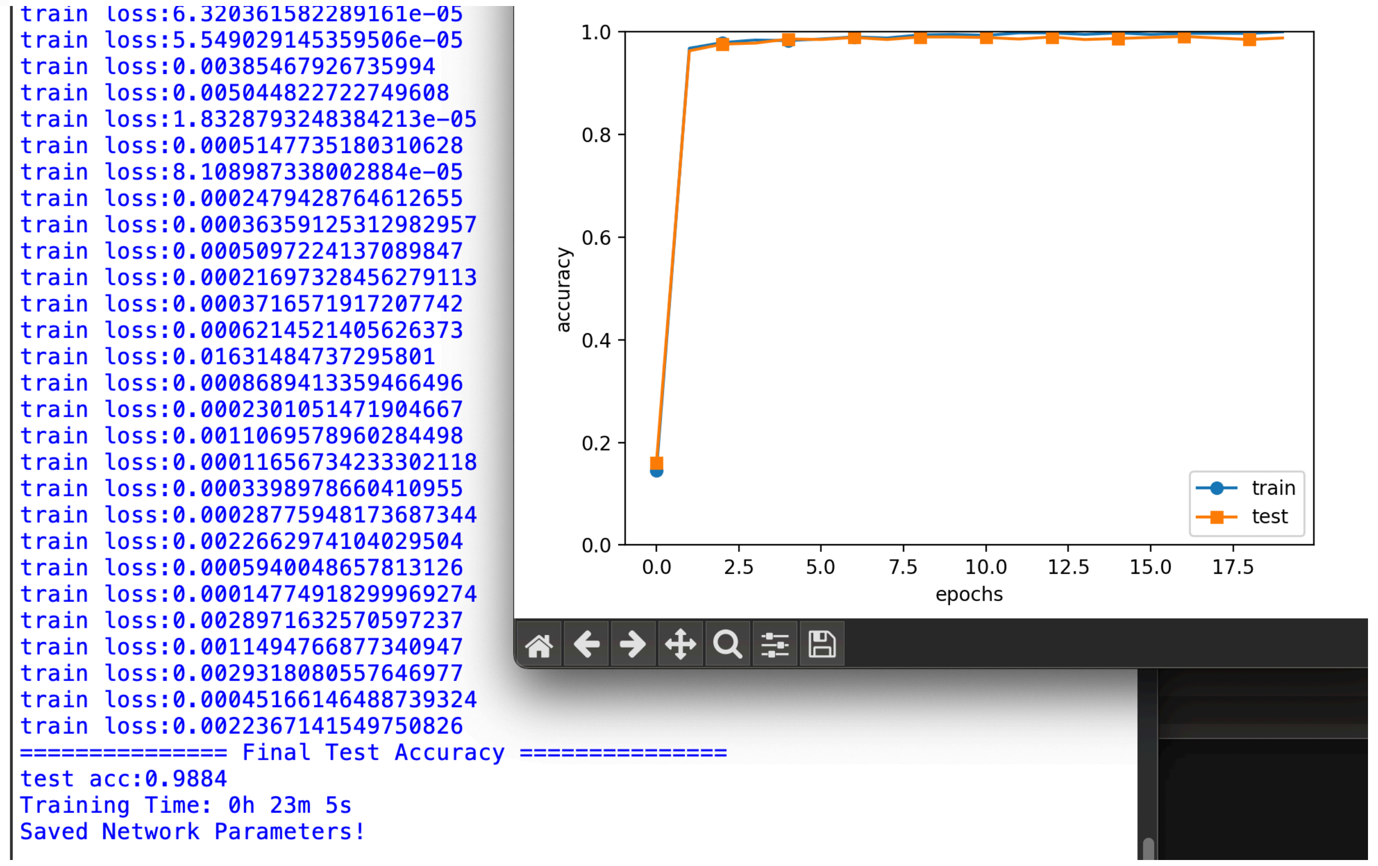






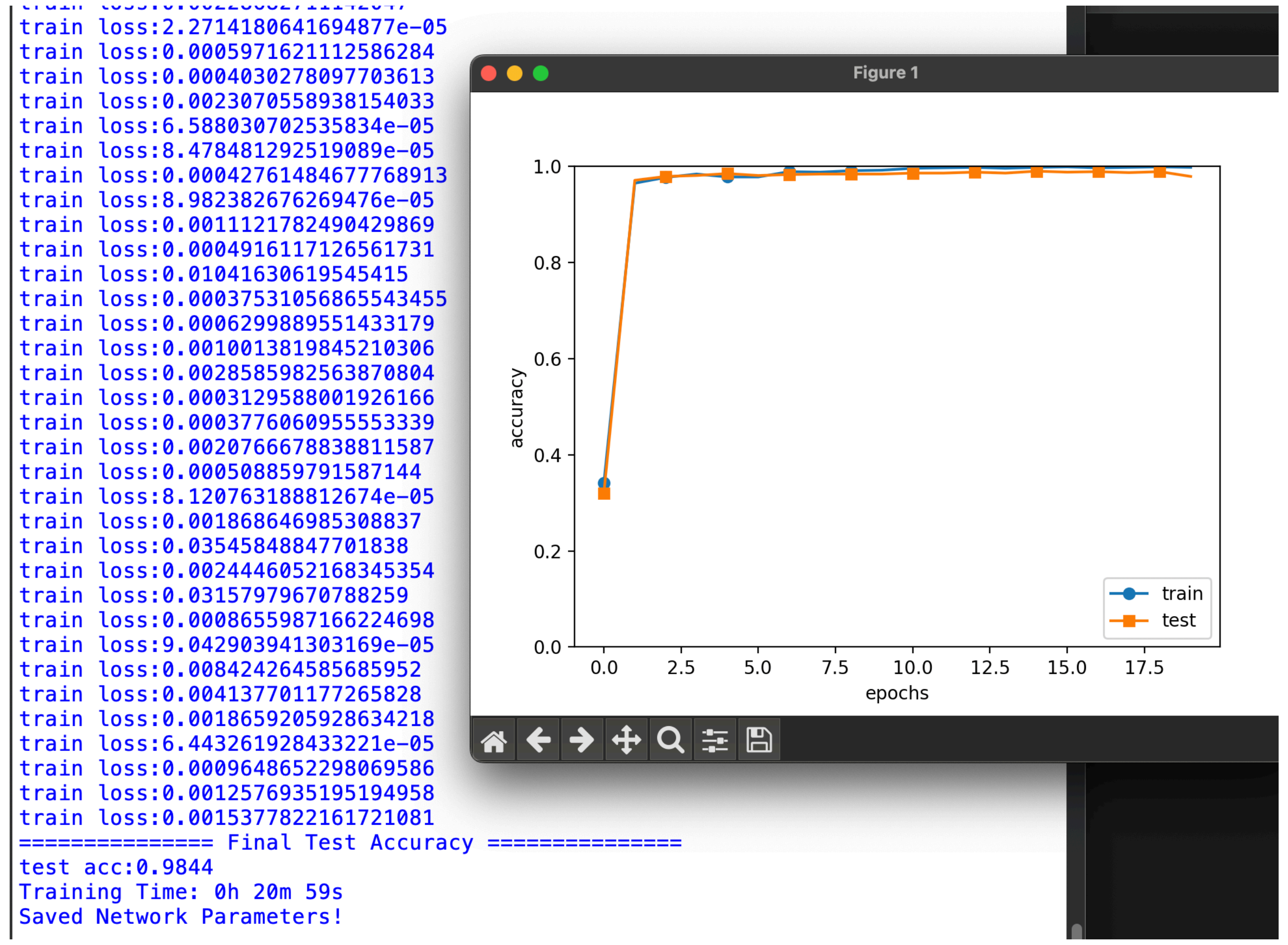


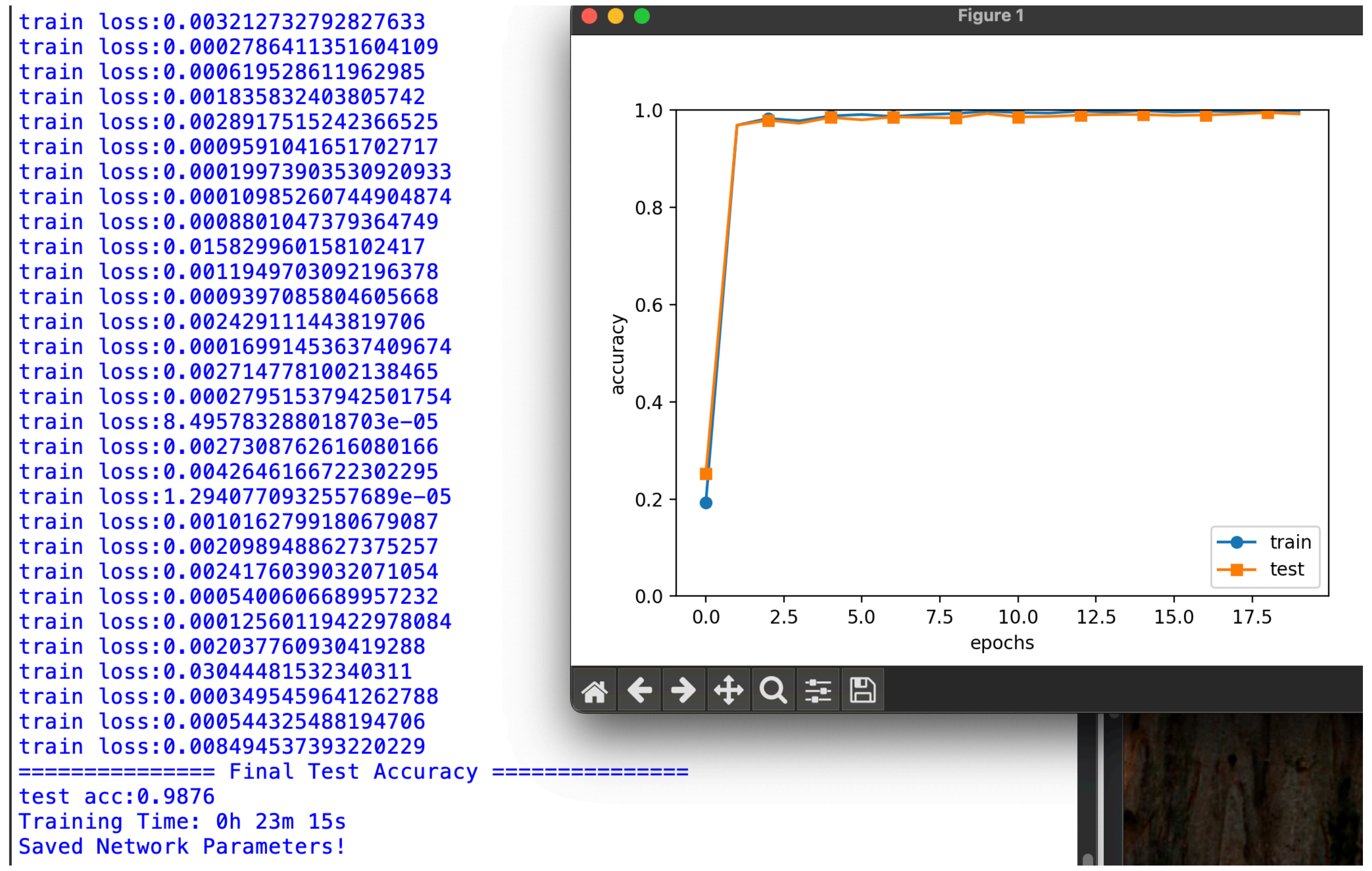


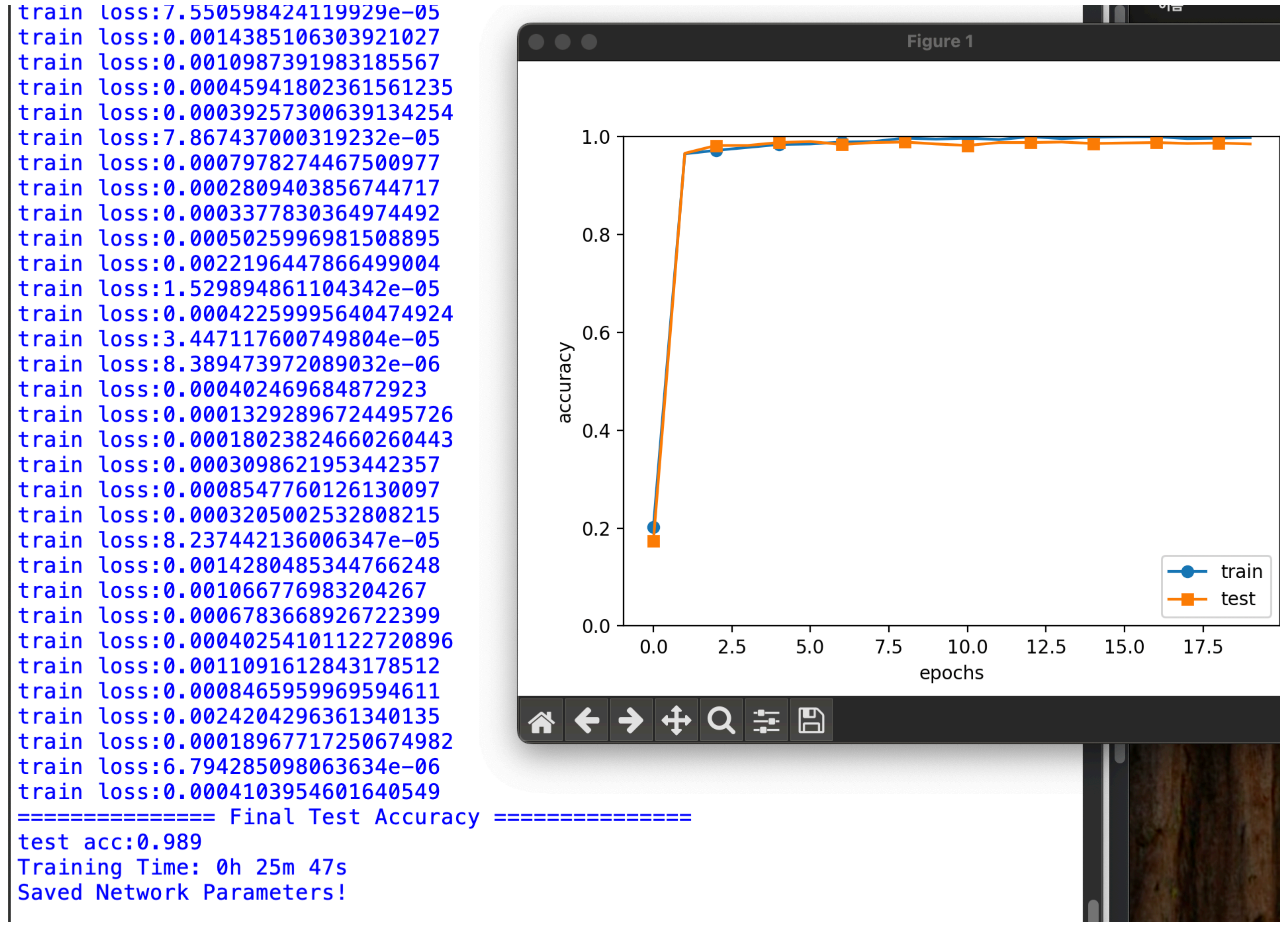


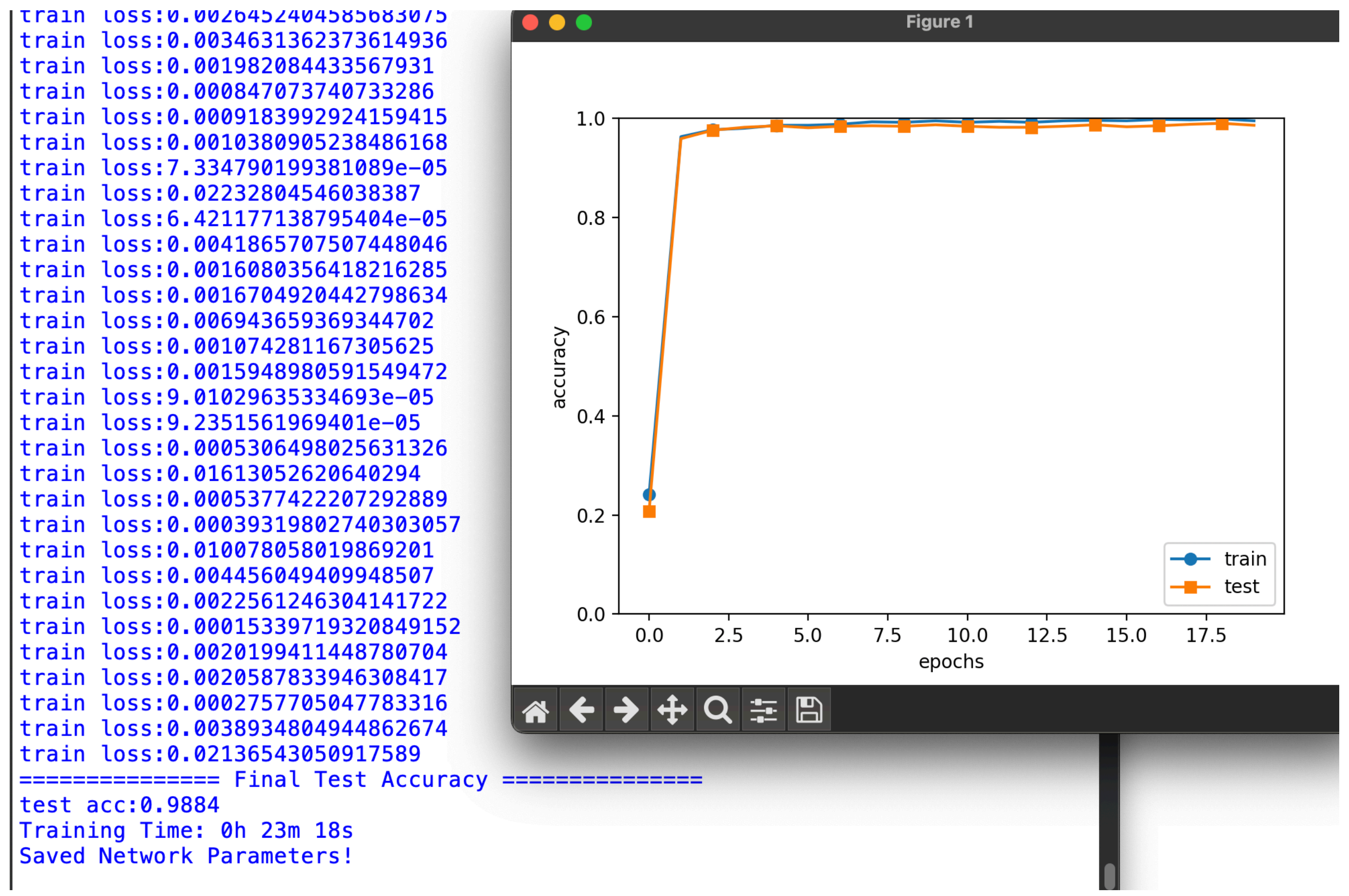
Chapter 07

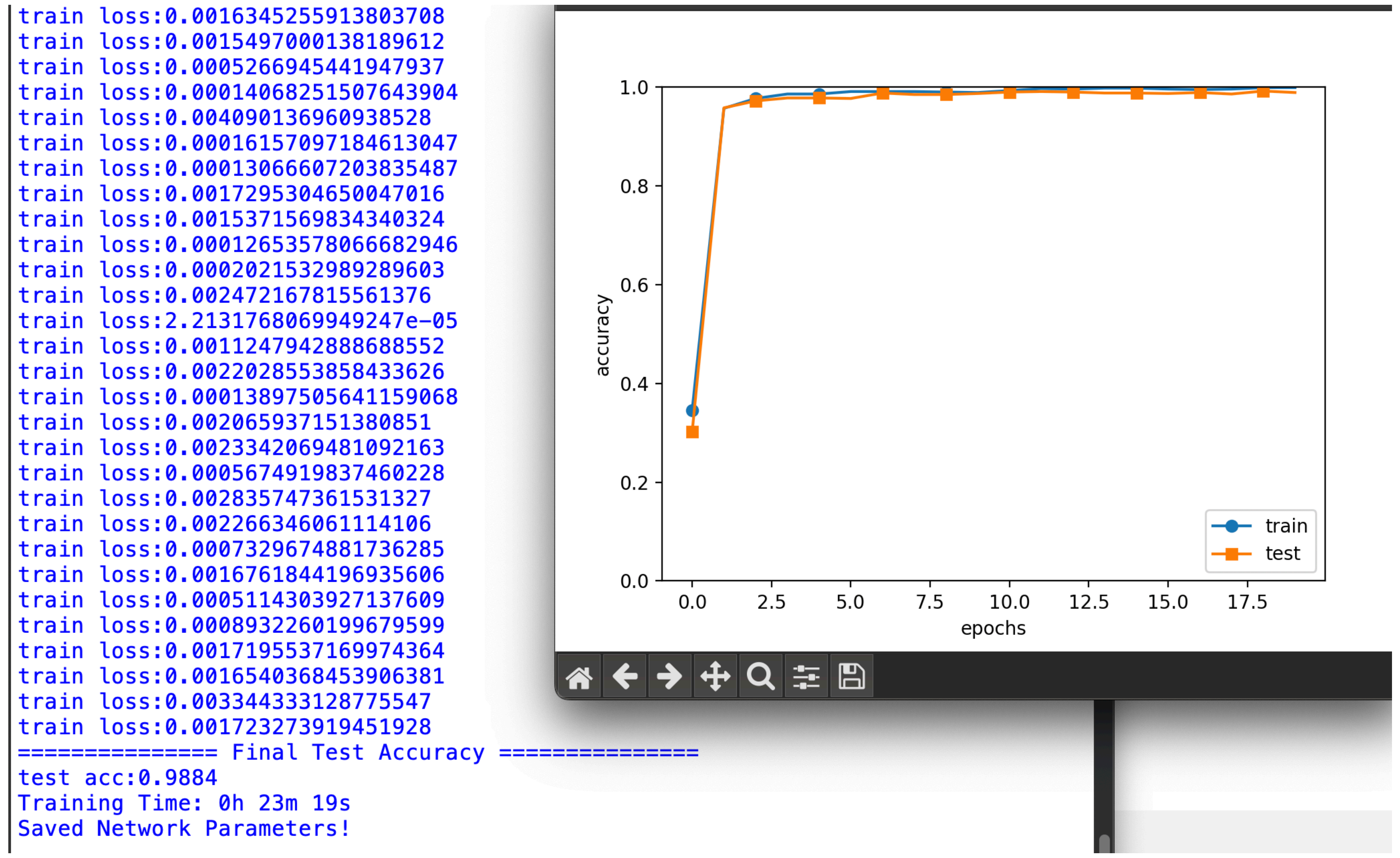
CNN 구현 - 교재 결과











MLP(Multi-Layer Perceptron, 다층 퍼셉트론)

신경망 모델

MLP: 여러 개의 뉴런이 층(layer) 형태로 연결된 신경망 모델

구조: 모든 뉴런이 다음 층의 모든 뉴런과 연결되는 구조 -> 완전연결층

특징: 1. 1차원 데이터(벡터) 입력으로 받음 -> 다차원 데이터는 1차원 형태로 변환 사용

2. 비선형 변환 가능 -> 활성화 함수를 통해 복잡한 비선형 패턴 학습 가능

학습과정: 순전파 -> 오차 계산 -> 역전파 -> 가중치 업데이트

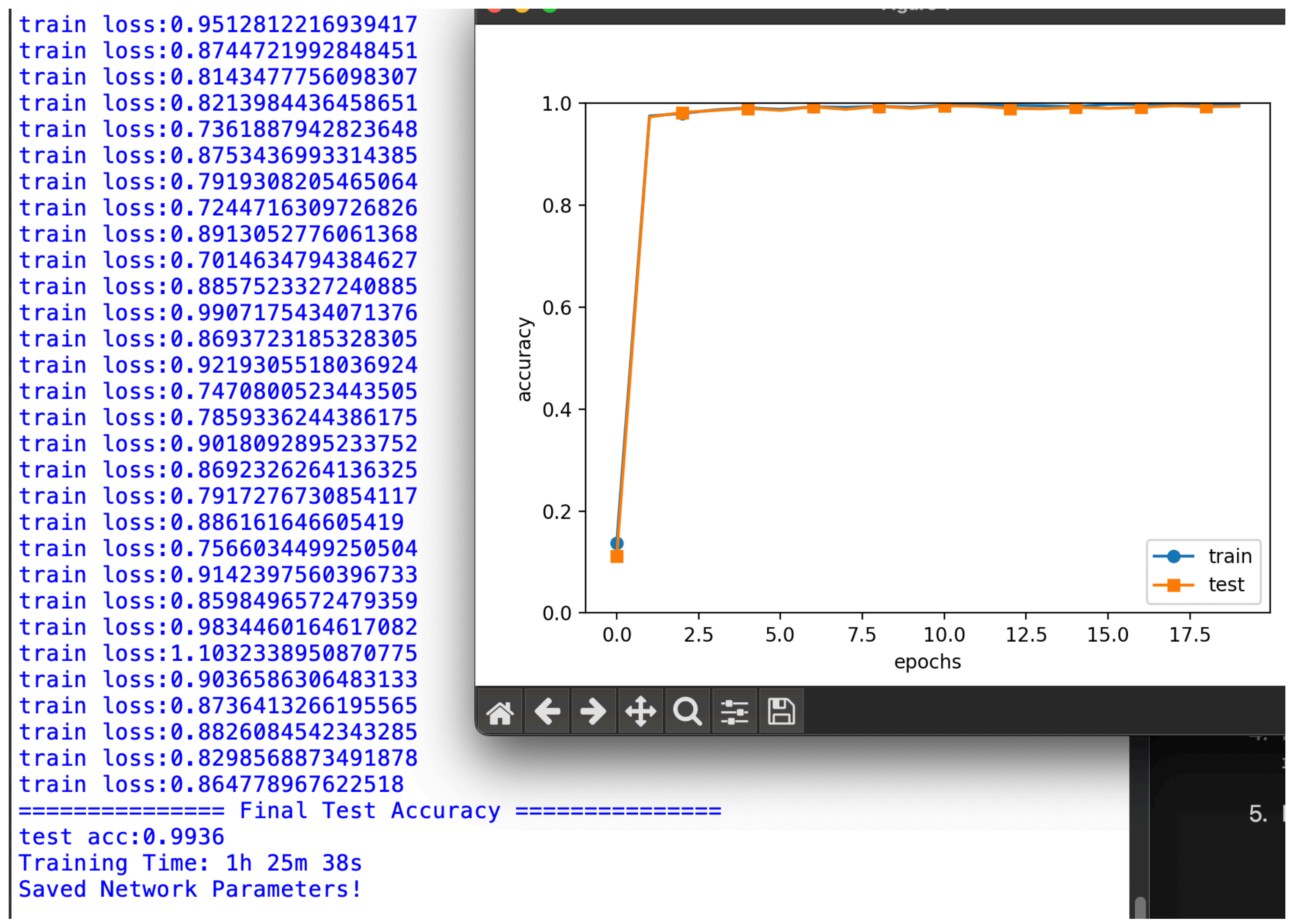
장점: 구조가 간단하고 구현이 쉬움, 어떤 유형의 데이터에도 적용 가능

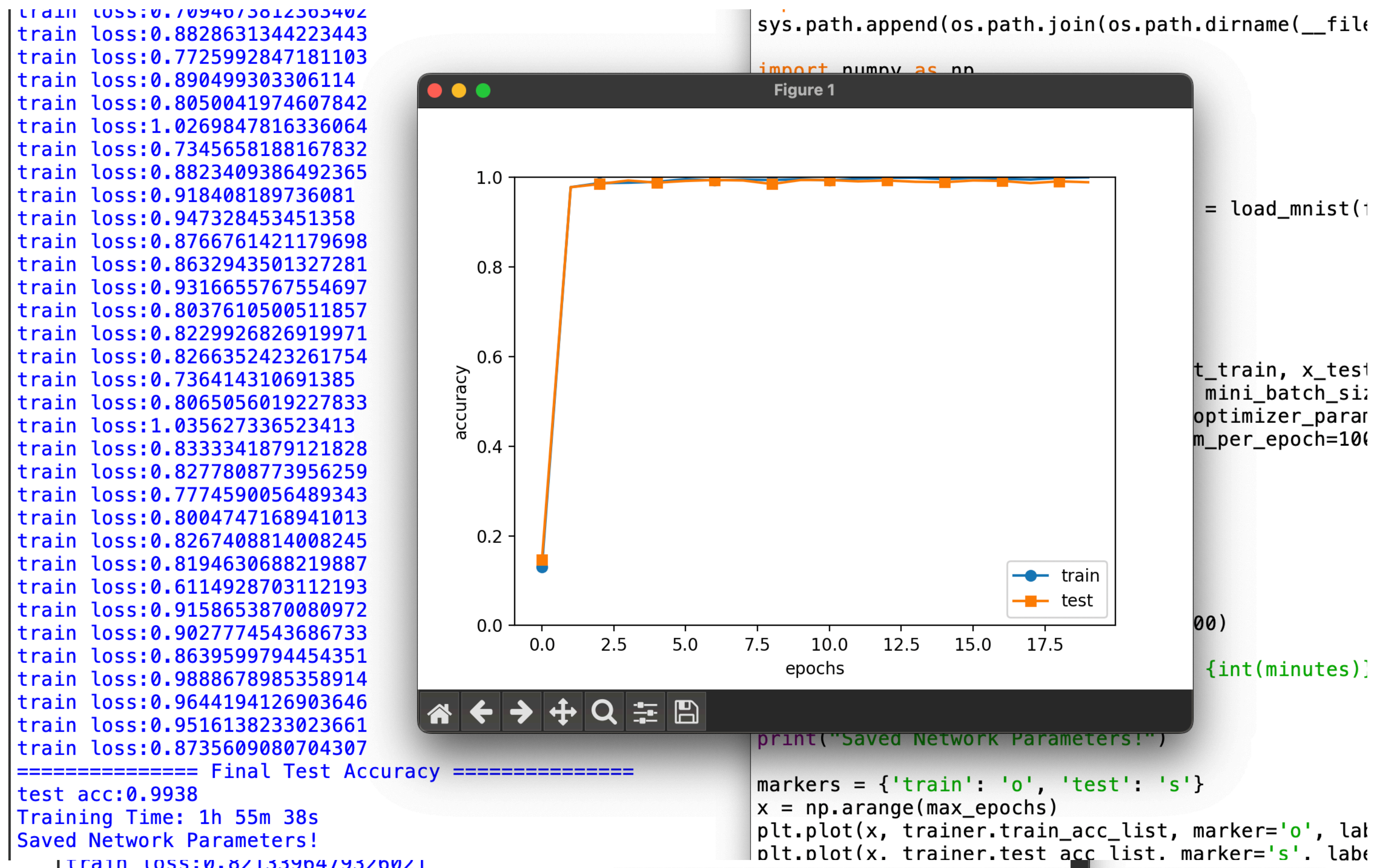
한계: 1. 고차원 데이터 처리에 비효율적 -> 공간적 구조를 학습할 수 없음

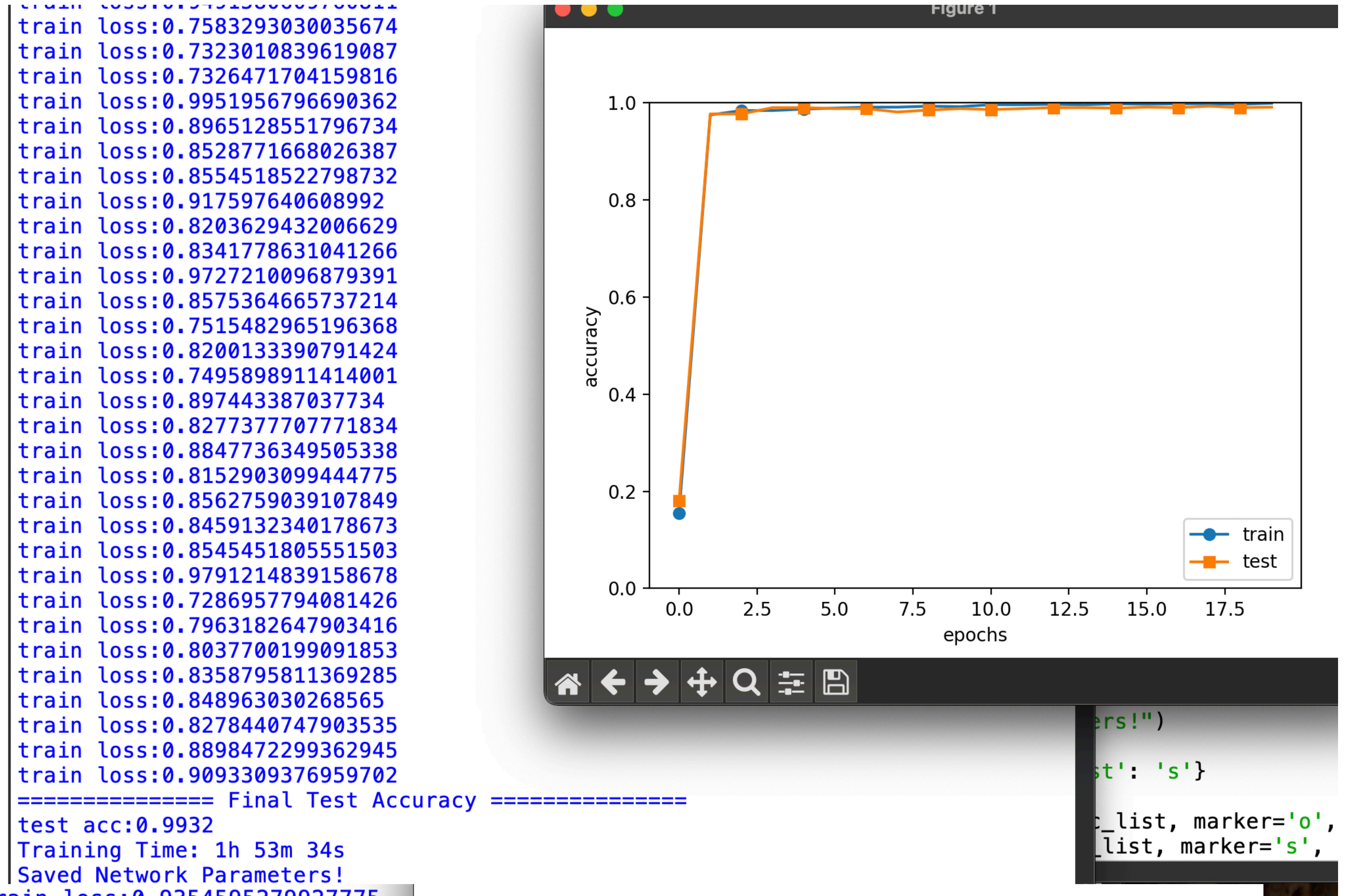
2. 가중치가 많아 연산량이 많음, 은닉층이 깊어질수록 학습속도 느림

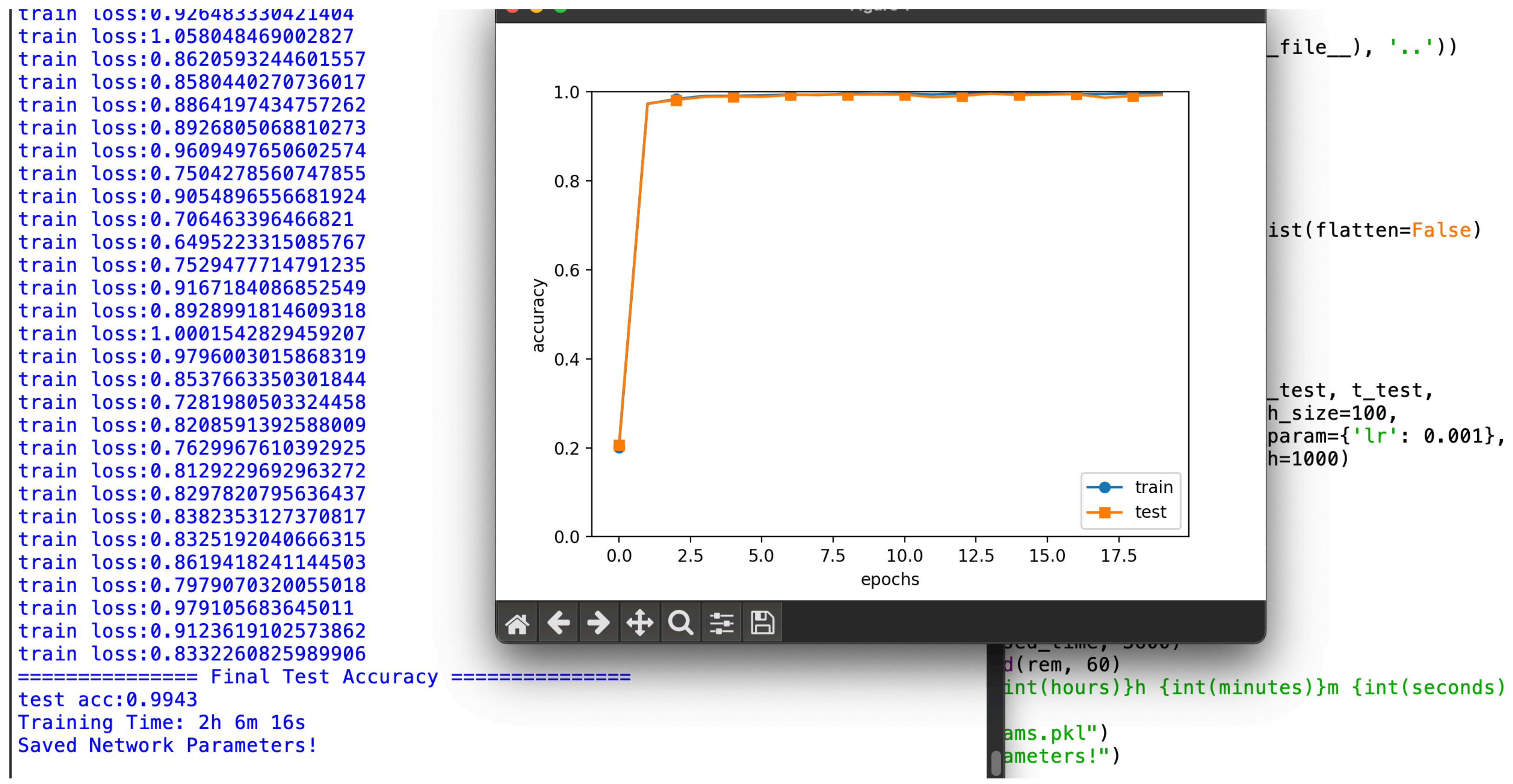
Chapter 08

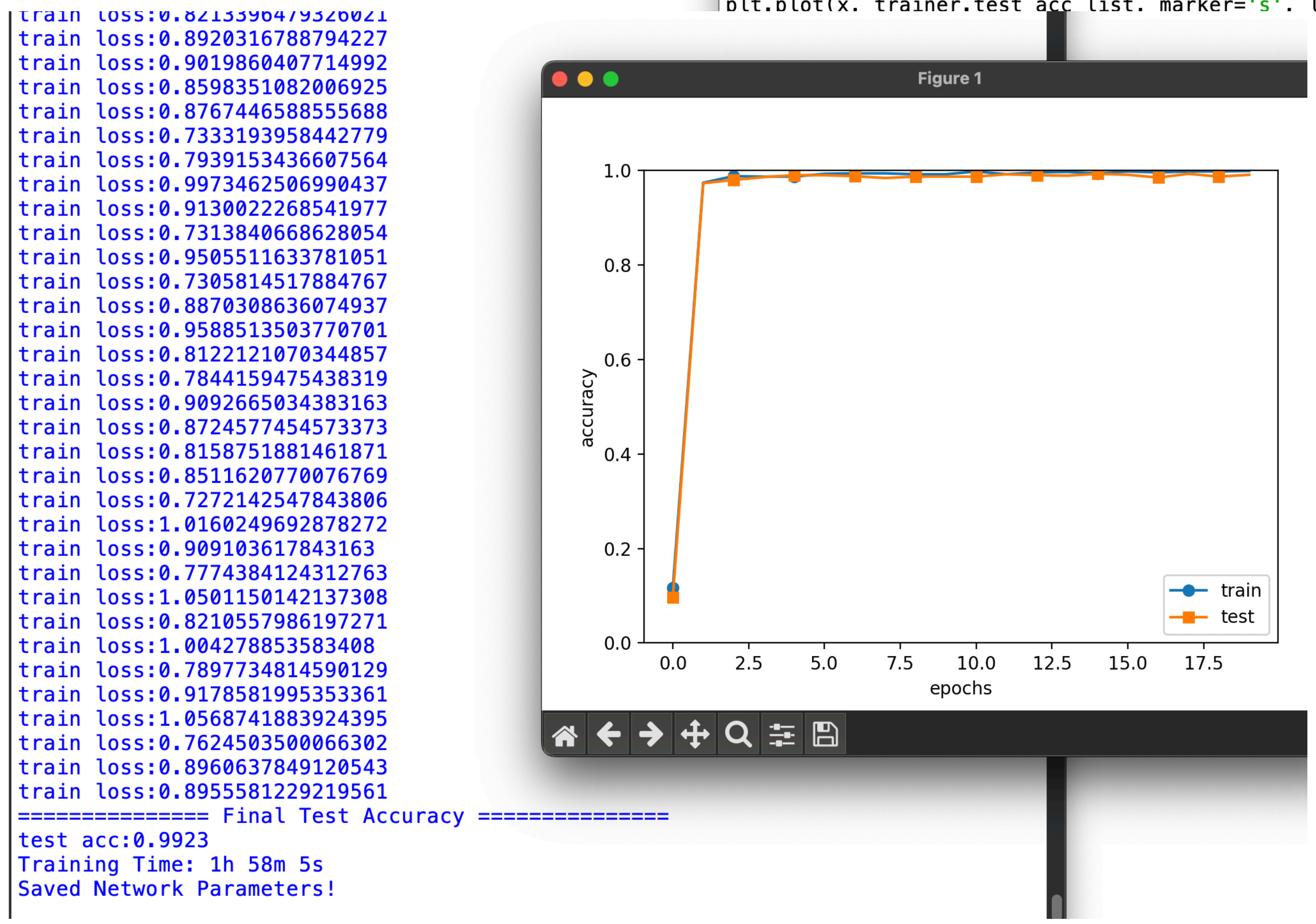
심층 신경망 구현 - 구현 결과

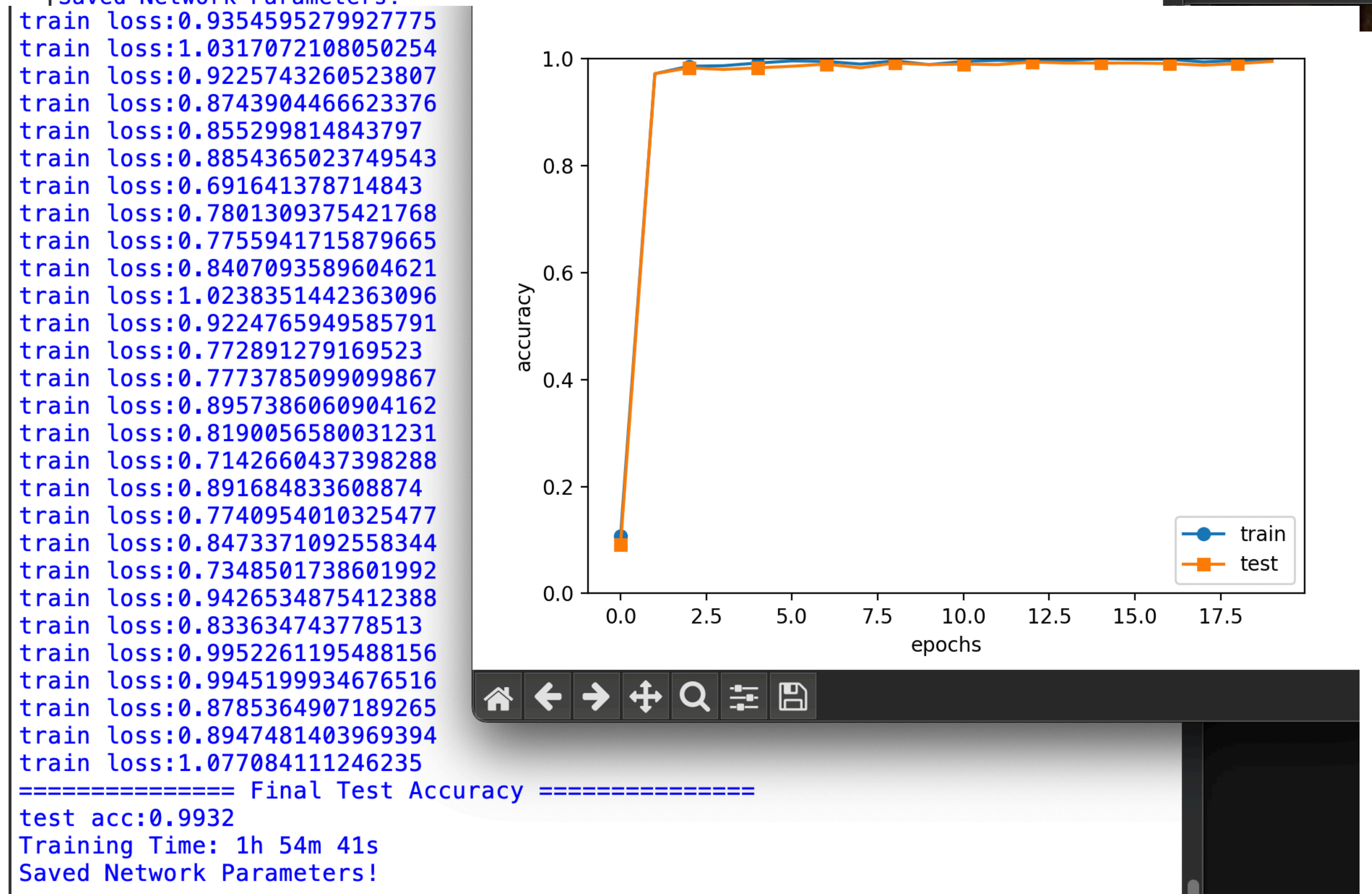








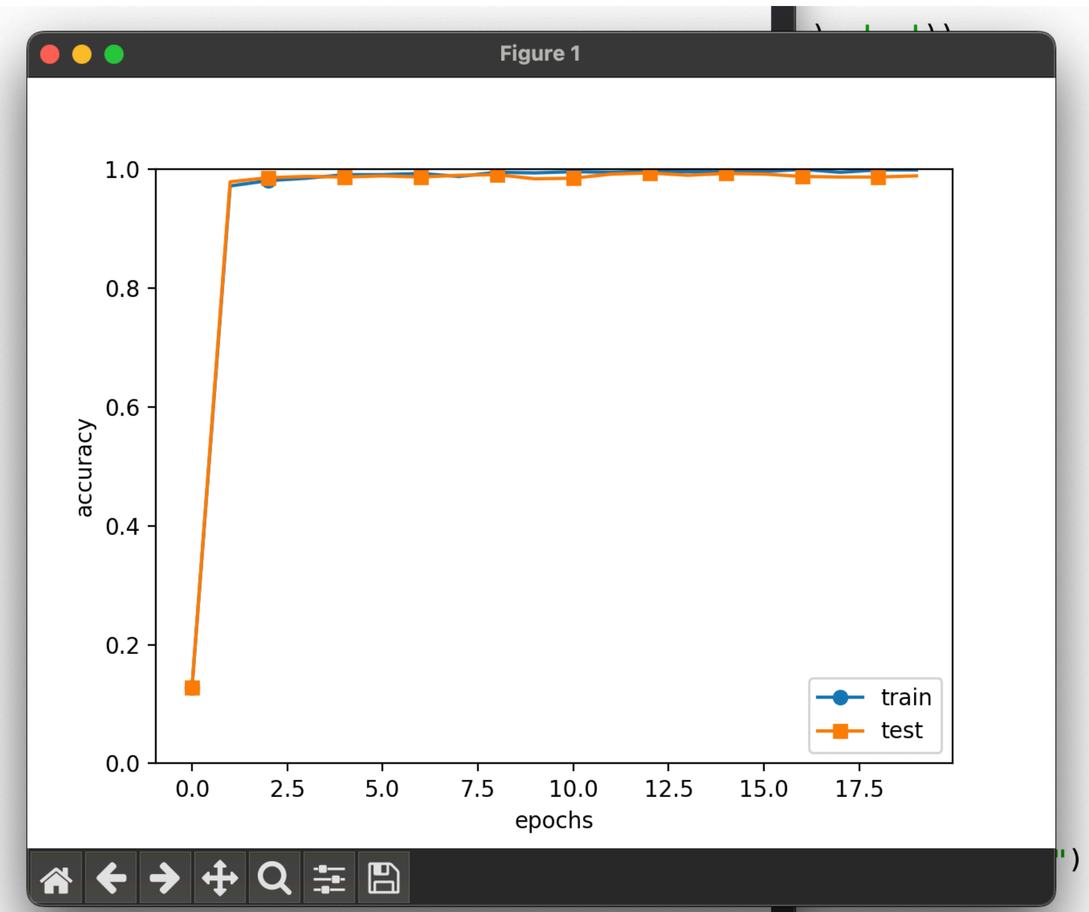




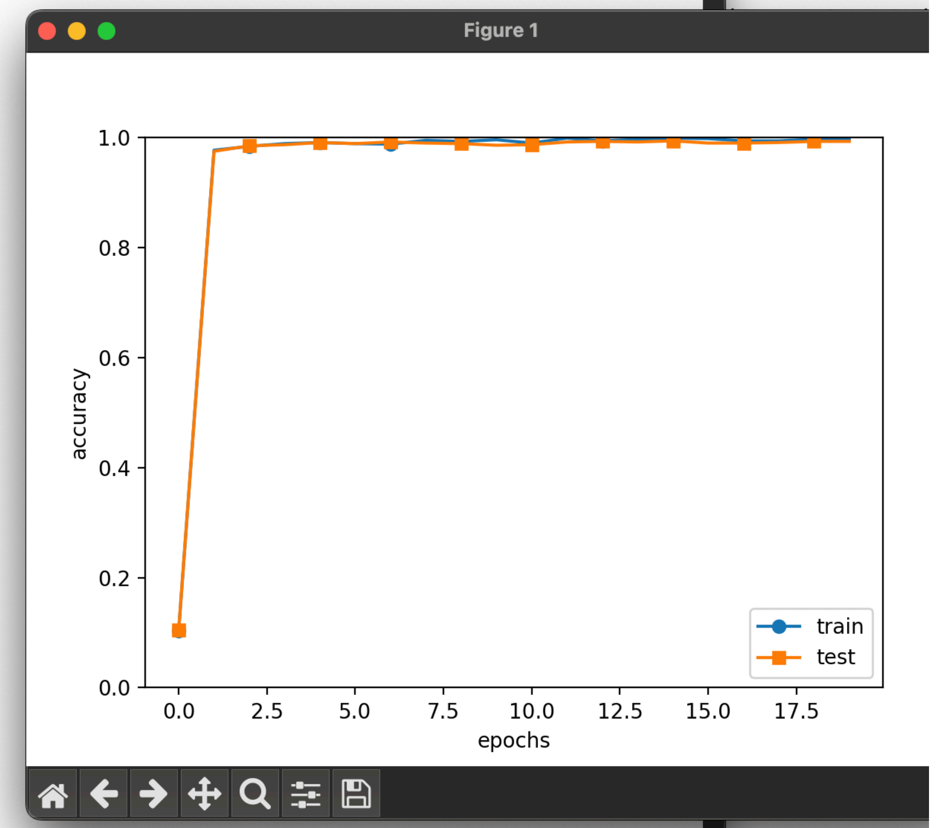
Chapter 08

심층 신경망 구현 - 교재 결과

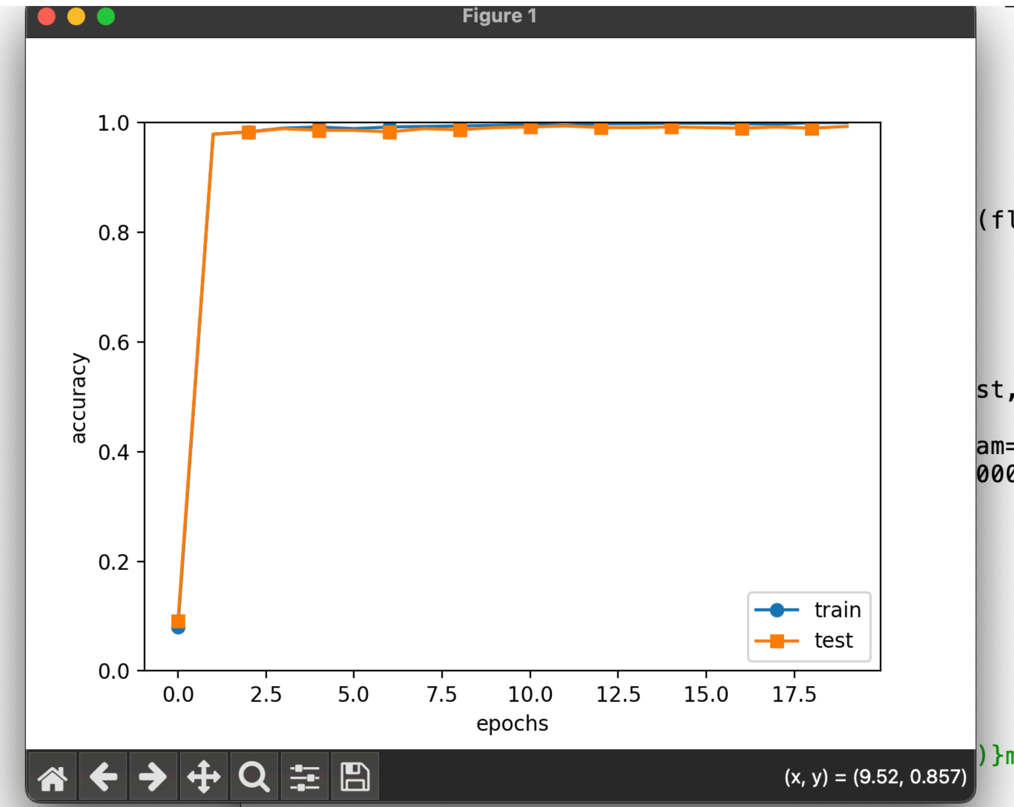
```
train loss:0.8296374858000141
train loss:0.7825392955204911
train loss:0.8116174386062852
train loss:1.085749029568787
train loss:0.8546250239447428
train loss:0.8427354169253489
train loss:0.9562240422452064
train loss:0.7320062556759582
train loss:0.764203698812868
train loss:0.7997293423765366
train loss:0.8474352606865421
train loss:0.9234552432698503
train loss:0.7551574040126969
train loss:0.9602282345364571
train loss:0.7608668672842023
train loss:0.8588898023480543
train loss:0.9579152414161785
train loss:0.9214805909033704
train loss:0.8105779973319658
train loss:0.8828376820352486
train loss:0.733813077005694
train loss:0.8070472993280333
train loss:0.7868187062924997
train loss:0.9226568120715612
train loss:0.8097551603778531
train loss:1.0993124908167875
train loss:0.8033242141801238
train loss:0.922829391763154
train loss:0.9167282390109031
train loss:0.747099158124897
train loss:0.9486248167803933
=============== Final Test Accuracy ===============
test acc:0.9938
Training Time: 2h 12m 53s
Saved Network Parameters!
```




```
train loss:0.8086517843632042
train loss:0.8164795839757152
train loss:0.7887283017065774
train loss:0.7464404018296132
train loss:1.0317397433181834
train loss:0.7112008953982147
train loss:0.8160721593570109
train loss:0.7427178768908747
train loss:0.874131177529602
train loss:1.0173317631878538
train loss:0.9367249281111825
train loss:0.9141656794807688
train loss:0.9485820561839334
train loss:0.8565692209408443
train loss:0.9098811188403698
train loss:0.7110979388396281
train loss:0.9593668675217815
train loss:0.9062695471597374
train loss:0.8018043224173845
train loss:0.6380737581273856
train loss:0.8617813219821981
train loss:0.8293763940357178
train loss:0.9054225390635371
train loss:0.6744271785934979
train loss:0.9341809114774741
train loss:0.8354875742863045
train loss:0.8793295551533921
train loss:0.8405170857434706
train loss:0.9177637294940072
train loss:0.7879983025228945
train loss:0.9375545572886708
train loss:0.7433432704815182
train loss:0.6731121927865934
=============== Final Test Accuracy ===============
test acc:0.9934
Training Time: 2h 21m 8s
Saved Network Parameters!
```



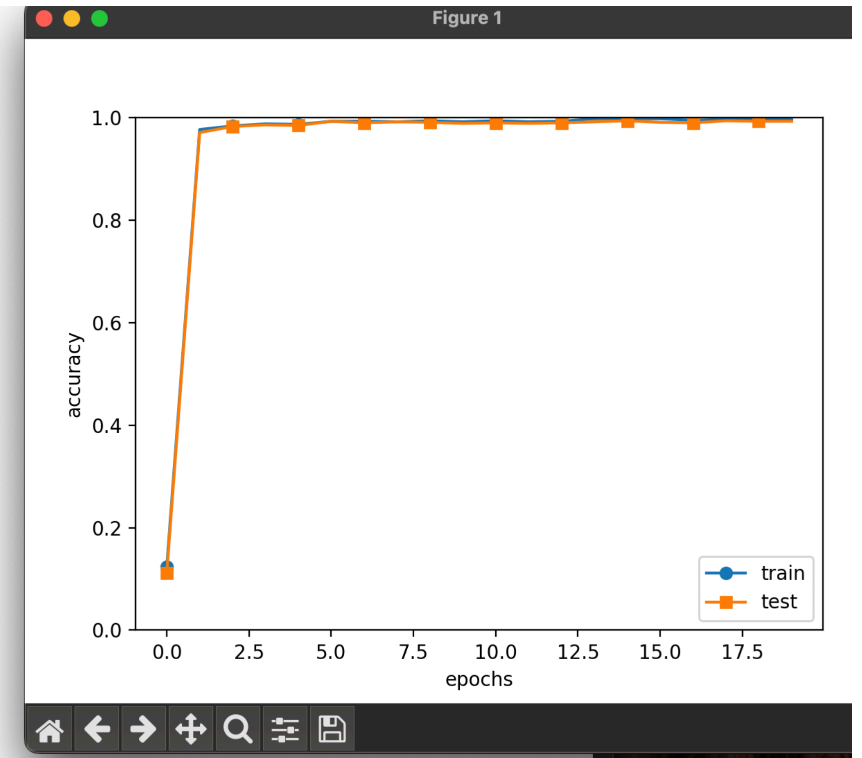

```
train loss:1.0019932891679237
train loss:0.9563523714903451
train loss:0.8754076749378406
train loss:1.0042968118622646
train loss:0.7435835546641428
train loss:1.013805305847344
train loss:0.8291285562736798
train loss:0.6628137108827272
train loss:0.8069435076155075
train loss:1.0071188608308466
train loss:0.9348147960626361
train loss:1.0048332908567013
train loss:0.924899513203548
train loss:0.9742784824691887
train loss:0.8372194278062456
train loss:0.7539691724850879
train loss:0.9292869008556714
train loss:0.8314039017381165
train loss:0.9385311009202549
train loss:0.9428479890272231
train loss:0.7884606164032035
train loss:0.9663536704143573
train loss:0.8425808691990937
train loss:0.8251935234592521
train loss:0.8439652878514866
train loss:0.8033822085362118
train loss:0.9526245820325401
train loss:0.8887144461775737
train loss:0.8784038276191859
train loss:0.8448889289285312
train loss:0.8484427930637668
=============== Final Test Accuracy ===============
test acc:0.9912
Training Time: 2h 3m 41s
Saved Network Parameters!
```



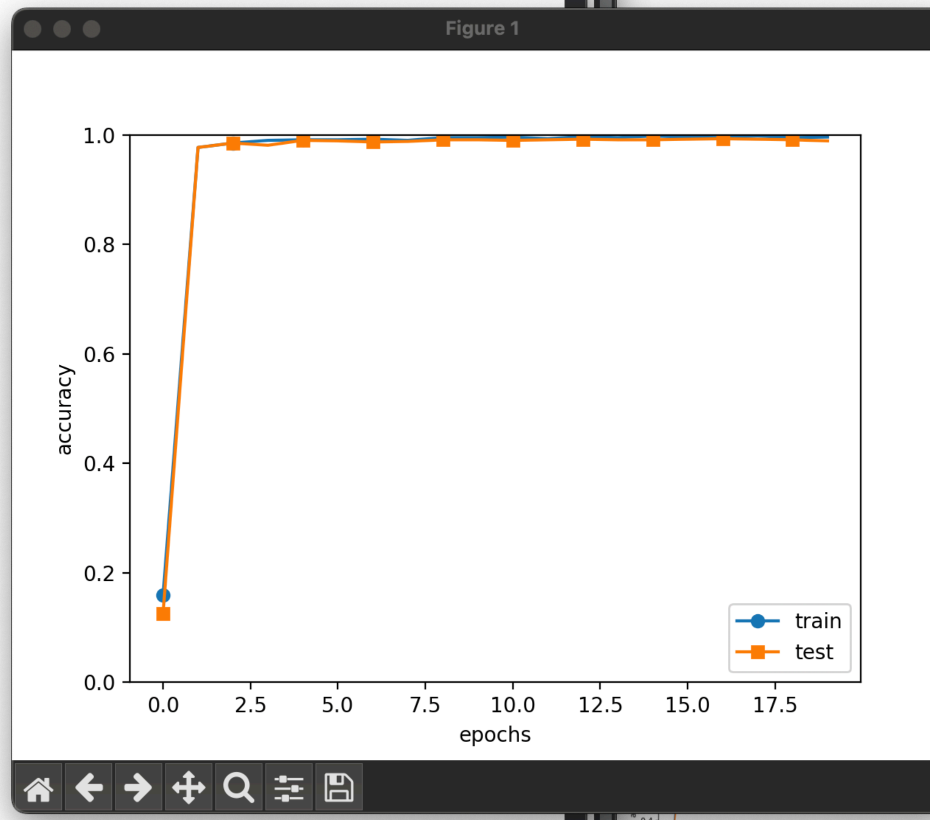

```
# 매개변수 보관
network.save_params("deep_convnet_params.pkl")
print("Saved Network Parameters!")

markers = {'train': 'o', 'test': 's'}
x = np.arange(max_epochs)
```

```
train loss:0.9426144757588321
train loss:0.7735305403397597
train loss:0.972521307660231
train loss:1.0131468214074022
train loss:0.872520984460602
train loss:0.7934414344355577
train loss:0.8946524982308055
train loss:0.9427004585488696
train loss:0.7842032972949915
train loss:0.97163411538311
train loss:0.818997875234948
train loss:0.8341111771556322
train loss:0.8938465697577976
train loss:0.8962286231812528
train loss:0.9339796113370668
train loss:0.9843426872514328
train loss:0.783549617984252
train loss:0.9845157135591478
train loss:1.0856430336041643
train loss:0.8760388553581429
train loss:0.8071513881587741
train loss:0.9117088627015322
train loss:0.7187288748219081
train loss:0.8839583647845346
train loss:0.9248462947816287
train loss:0.6745672860834362
train loss:0.9167824144271727
train loss:0.8529746770820346
train loss:1.0334101465447791
train loss:1.027388349950858
train loss:0.9061032612596405
=============== Final Test Accuracy ===============
test acc:0.9947
Training Time: 1h 54m 2s
Saved Network Parameters!
```




```
train loss:0.8731653516059842
train loss:0.9025188786104096
train loss:0.8052048431451717
train loss:0.7619775540029106
train loss:0.6856544164045942
train loss:0.8162179830985146
train loss:0.8643439854820365
train loss:0.9599603532940059
train loss:0.7562935252017223
train loss:0.7817565503365764
train loss:0.8626108909020153
train loss:1.0187581052115575
train loss:0.7824294724623977
train loss:0.9537337248865723
train loss:0.8415014164332172
train loss:0.912437443197851
train loss:0.8958508607537784
train loss:0.8845983782575612
train loss:1.0207557300483663
train loss:0.861387769595567
train loss:0.6780542291657885
train loss:0.8952419142170469
train loss:0.8490771294397096
train loss:0.837051065293601
train loss:0.8007098606647425
train loss:0.8752064014720871
train loss:0.8416984770888679
train loss:0.8573161371331115
train loss:0.8177264884235007
train loss:0.8419197240428115
train loss:0.8484535274571732
train loss:0.8205551963674432
=============== Final Test Accuracy ===============
test acc:0.9939
Training Time: 1h 53m 37s
Saved Network Parameters!
```




```
train loss:0.768341641816467
train loss:0.8320452106872918
train loss:0.9512524167754904
train loss:0.9387508417778545
train loss:1.0346964622611983
train loss:0.9028641241925734
train loss:0.7579355560452631
train loss:0.739005414456749
train loss:0.8479864036940448
train loss:0.8384192732679723
train loss:0.8780525702300269
train loss:0.8781489867317716
train loss:0.7949731274704347
train loss:0.8889400132589845
train loss:0.9784252277164167
train loss:0.934695485377691
train loss:0.7292497652948126
train loss:0.8109726293447431
train loss:0.7514849569642361
train loss:0.6879678036271697
train loss:0.8753052342621396
train loss:0.792253734937181
train loss:0.7253938186123037
train loss:0.8251628394763031
train loss:0.8903776612991505
train loss:0.9457188587891722
train loss:0.730725319310627
train loss:0.9479442577534201
train loss:0.9329350139762395
train loss:0.9053611826457533
=============== Final Test Accuracy ===============
test acc:0.9915
Training Time: 1h 53m 48s
Saved Network Parameters!
```